Distance Learning Centre

Ahmadu Bello University

Zaria, Nigeria

LIBS 894:
Information Retrieval

Course Material

Programme: Masters in Information Management

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QUOTE

The retrieval process begins when a lack of information shows itself in a human mind and the decision is taken to find out if this information has been discovered and published
Douglas John Foskett (1963)

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Course Study Guide

Course Information

Course Code: LIBS 894
Course Title: Information Retrieval
Credit Units: 2
Semester: Second

Course Introduction and Description

Introduction:

You are welcome to LIBS894, this is a 2credit unit year one second semester course. I will be your guide in the span of this course, you should feel free to ask questions when in a form of difficulties relating this course, and my door is always open.

Description:

The course should engage you for about 15 weeks in a semester. It is designed to provide reading material for two to three hours of study. This course does not require prior skills in any area of knowledge other than the general admission requirements. The approach at this level is to help the students to appreciate the basic characteristics of information, the need for good **organisation of information**, and the fundamental concepts of **storage and retrieval**.

The rest of this guide will tell you what you are expected to learn in this course, how to work through the course, the content of the course, and useful information on exercises and assignments, and how to get the most out of the course.

i. COURSE PREREQUISITES

You should note that although this course has no subject pre-requisite, you are expected to have:

1. Satisfactory level of English proficiency
2. Basic Computer Operations proficiency
3. Online interaction proficiency
4. Web 2.0 and Social media interactive skills

ii. COURSE LEARNING RESOURCES

i. Course Textbooks

1. Baeza-Yates, R. and Ribeiro-Neto, B. (1999). *Modern Information Retrieval*.Addison-Wesley
2. Chowdhury, G.G (2003). *Introduction to Modern Information Retrieval*.Neal-Schuman
3. Jones, K. S. & Willett, P. (1997). *Readings in Information Retrieval*. Morgan Kaufmann.
4. Kowalski, G. &Maybury, M.T. (2005).*Information Storage andRetrieval Systems*.Springer.
5. van Risjbergen, C.J. (2004). *The Geometry of Information Retrieval*.Cambridge UP.

iii. COURSE OUTCOMES

After studying this course, you should be able to:

1. Explain the concepts of data, data processing, and information.
2. Distinguish between document and information and describe the processes of documentation.
3. Correctly assign key words to be used for retrieval purposes.
4. Describe the basic architecture of a computer and the role of computers in

storage and retrieval.

5. Describe the storage media in use and the basic structure of records, files, and databases.
6. Explain the concepts of information retrieval.
7. Identify the information retrieval components.
8. Identify the Information Retrieval Models.
9. Discuss user characteristics and user needs, which are fundamental to information storage and retrieval.
10. Negotiate Queries using Reference Interview.
11. Describe different Methods of Querying.
12. Correctly analyse requests for information and formulate search strategies.
13. Retrieve information from an Information System.
14. Identify the factors the affect online search.
15. Optimize retrieval process.
16. Evaluate retrieval product.
17. Search the Internet.

iv. ACTIVITIES TO MEET COURSE OBJECTIVES

To go through this course you are required to read the study units, answer the self-assessment exercises and do the assignment in each unit. The self-assignment exercises are meant to help you to reinforce what you have learnt. It will be very helpful to you to try and answer the questions first before looking at the answers.

At the end of the unit, you will find an assignment, which will be marked by your tutor. Work diligently on it and submit your work to your tutor for grading.

Individual assignments/test, Group assignments, Discussions/Quizzes/Out of class engagements etc will constitute 40% of the total marks of the examination in this course.

You will need to have access to a computer and be familiar with the basic elements of a computer system. In due course, you will have practical exercises in the

library and on the Internet, and so, you need to have access to these facilities. It is expected that you will spend on the average two to three hours to study one unit and about 15weeks to complete the whole course. However, you should realise that you are actually to work at your own pace.

Specifically, this course shall comprise of the following activities:

1. Studying courseware
2. Listening to course audios
3. Watching relevant course videos
4. Field activities, industrial attachment or internship, laboratory or studio work (whichever is applicable)
5. Course assignments (individual and group)
6. Forum discussion participation
7. Tutorials (optional)
8. Semester examinations (CBT and essay based).

v. TIME (TO COMPLETE SYLABUS/COURSE)

To cope with this course, you would be expected to commit a minimum of 3 hours Weekly for the Course.

vi. GRADING CRITERIA AND SCALE

Grading Criteria

A. Formative assessment

Grades will be based on the following:

Individual assignments/test (CA 1,2etc.)	20
Group assignments (GCA 1, 2 etc.)	10
Discussions/Quizzes/Out of class engagements etc.	10

B. Summative assessment (Semester examination)

CBT based	30
Essay based	30
TOTAL	**100%**

C. Grading Scale:

A = 70-100
B = 60 – 69
C = 50 - 59
D = 45-49
F = 0-44

D. Feedback

Courseware based:

1. In-text questions and answers (answers preceding references)
2. Self-assessment questions and answers (answers preceding references)

Tutor based:

1. Discussion Forum tutor input
2. Graded Continuous assessments

Student based:

1. Online programme assessment (administration, learning resource, deployment, and assessment).

vii. LINKS TO OPEN EDUCATION RESOURCES

OSS Watch provides tips for selecting open source, or for procuring free or open software.

SchoolForge and SourceForge are good places to find, create, and publish open software. SourceForge, for one, has millions of downloads each day.

Open Source Education Foundation and Open Source Initiative, and other organisation like these, help disseminate knowledge.

Creative Commons has a number of open projects from Khan Academy to Curriki where teachers and parents can find educational materials for children or learn about Creative Commons licenses. Also, they recently

launched the School of Open that offers courses on the meaning, application, and impact of "openness."

Numerous open or open educational resource databases and search engines exist. Some examples include:

- OEDb: over 10,000 free courses from universities as well as reviews of colleges and rankings of college degree programmes
- Open Tapestry: over 100,000 open licensed online learning resources for an academic and general audience
- OER Commons: over 40,000 open educational resources from elementary school through to higher education; many of the elementary, middle, and high school resources are aligned to the Common Core State Standards
- Open Content: a blog, definition, and game of open source as well as a friendly search engine for open educational resources from MIT, Stanford, and other universities with subject and description listings
- Academic Earth: over 1,500 video lectures from MIT, Stanford, Berkeley, Harvard, Princeton, and Yale
- JISC: Joint Information Systems Committee works on behalf of UK higher education and is involved in many open resources and open projects including digitising British newspapers from 1620-1900!

Other sources for open education resources

Universities

- The University of Cambridge's guide on Open Educational Resources for Teacher Education (ORBIT)
- OpenLearn from Open University in the UK

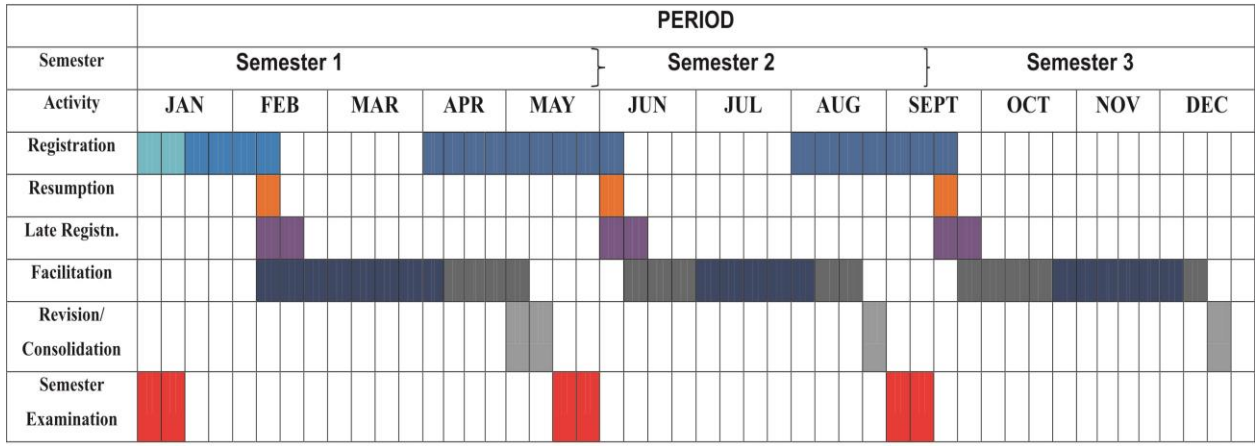
Global

- Unesco's searchable open database is a portal to worldwide courses and research initiatives
- African Virtual University (http://oer.avu.org/) has numerous modules on subjects in English, French, and Portuguese
- https://code.google.com/p/course-builder/ is Google's open source software that is designed to let anyone create online education courses

- Global Voices (http://globalvoicesonline.org/) is an international community of bloggers who report on blogs and citizen media from around the world, including on open source and open educational resources

  Individuals (which include OERs)
- Librarian Chick: everything from books to quizzes and videos here, includes directories on open source and open educational resources
- K-12 Tech Tools: OERs, from art to special education
- Web 2.0: Cool Tools for Schools: audio and video tools
- Web 2.0 Guru: animation and various collections of free open source software
- Livebinders: search, create, or organise digital information binders by age, grade, or subject (why re-invent the wheel?)

ABU DLC ACADEMIC CALENDAR/PLANNER



N.B: - Semester Examinations 1st/2nd Week January

- All Sessions commence Mid-February
- 1 Week break between Semesters and 4 Weeks vocation at end of session.
- Semester 3 is OPTIONAL (Fast-tracking, making up carry-overs & deferments)

ix. COURSE STRUCTURE AND OUTLINE

Course Structure

WEEK	MODULE	STUDY SESSION	ACTIVITY
Week 1	**STUDY MODULE 1: Overview Of Information Retrieval**	**Study Session 1:** Data and Information ***Pp. 25***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site `https://bit.ly/2C14W1y`) 5. View referred Animation (Address/Site https://bit.ly/2HNFRZT)
Week 2		**Study Session 2:**Information Retrieval Definitions and Concepts ***Pp. 36***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube address `https://bit.ly/2NGfUOk`) 5. View referred Animation (Address/Site https://bit.ly/30DPN0C)
Week 3		**Study Session 3**: Information Retrieval Systems ***Pp. 45***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube address https://bit.ly/1bhYPOq) 5. View referred Animation (Address/Site https://bit.ly/30DPN0C)
Week 4		**Study Session 1:**Information Retrieval Models: Introduction	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session

	STUDY MODULE 2: Information Retrieval Models	***Pp. 54***	4. View any other Video/U-tubehttps://bit.ly/2Uk2WZo) 5. View referred Animation (Address/Site https://bit.ly/2Wk8WFx)
Week 5		**Study Session 2:**Boolean Modeland Extended Boolean Model ***Pp. 63***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/1hxT0ot) 5. View referred Animation (Address/Site `https://bit.ly/2YJyIk8`)
Week 6		**Study Session 3:**Vector Space Model and Fuzzy Model ***Pp. 71***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2TqRNsu) 5. View referred Animation (Address/Site `https://bit.ly/2HwVeai`)
Week 7		**Study Session 4:** Probabilistic Model and Natural Language Model ***Pp. 79***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2SGOJ79) 5. View referred Animation (Address/Site `https://bit.ly/2jHWJFQ`)
Week 8	**MODULE 3 Query and Query Negotiation**	**Study Session 1:** Query-an overview ***Pp. 88***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2tPPAs1) 5. View referred Animation (Address/Site https://bit.ly/30Ap7hv)

Week 9		**Study Session 2:** Query Structure ***Pp. 95***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2EJntRR) 5. View referred Animation (Address/Site https://bit.ly/30EtqbG)
Week 10		**Study Session 3:** Query Negotiation **Pp. 106**	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2Vtn6QO) 5. View referred Animation (Address/Site https://bit.ly/2HKpLAk)
Week 11	**MODULE 4: Information Retrieval Process And Evaluation**	**Study Session 1:** Information Retrieval Process-an overview ***Pp. 113***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2Vtn6QO) 5. View referred Animation (Address/Site https://bit.ly/30DPN0C)
		Study Session 2: Search Strategy ***Pp. 121***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/1FtyP54) 5. View referred Animation (Address/Site https://bit.ly/2JA0M5U)

Week 12		**Study Session 3:** Evaluation of Retrieval Products ***Pp. 129***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2tLZsTI) 5. View referred Animation (Address/Site https://bit.ly/2HQ9Osj)
		Study Session 4: Standard Metadata and their description ***Pp. 138***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube https://bit.ly/2TxoAfk https://bit.ly/2TvwqpO) 5. View referred Animation (Address/Site https://bit.ly/2HAn1Xy)
Week 13		**REVISION/TUTORIALS (On Campus or Online)& CONSOLIDATION WEEK**	
Week 14 & 15		**SEMESTER EXAMINATION**	

Course Outline

MODULE 1: Overview of Information Retrieval

Study Session 1: Data and Information
Study Session 2: Information Retrieval Definitions and Concepts
Study Session 3: Information Retrieval Systems

MODULE 2: Information Retrieval Models

Study Session 1: Information Retrieval Models- Introduction
Study Session 2: Boolean Model and Extended Boolean Model
Study Session 3: Vector Space Model and Fuzzy Model
Study Session 4: Probabilistic Model and Natural Language Model

MODULE 3: Query and Query Negotiation

Study Session 1: Query – an overview
Study Session 2: Query Structure
Study Session 3: Query Negotiation

MODULE 4: Information Retrieval Process

Study Session 1: Information Retrieval Process- an overview
Study Session 2: Search Strategy
Study Session 3: Evaluation Retrieval Products
Study Session 4: Standard Metadata and their description

Study Modules

Contents:

Study Session 1: Data and Information
Study Session 2: Concept of Information Retrieval
Study Session 3: Information Retrieval Systems

Study Session 1

Data and Information

Section and Subsection Headings:

Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Data
        2.1.1- Data Processing
    2.2- Information
        2.2.1- Value of Information
3.0Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 1, we will explore the concept of **data and information**, and you will learn how to use the term properly, because we have found out that many people use the tern wrongly and how to appreciate the value of **date and information**.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Explain the meaning of Data
2. Explain the concept of Data Processing
3. Explain the meaning of Information
4. Distinguish between Data and Information

2.0 Main Content

2.1 Data

What do you understand by the term data?

The word "**data**" is the plural form of the word "**datum**". **Data** may be regarded as symbols or figures that have potential value or to which meaning can be given. Now, let us consider how people record events. Think of the old man in the village who made nine strokes of chalk on the lintel of his front door to remind him that it was exactly nine hundred naira that he borrowed.

Fig 1.1.1: Data

Each time he paid back one hundred naira, he wiped away one stroke, when he was able to pay back a multiple of one hundred naira, he cleaned off the corresponding number of strokes. Would you not consider such a man as well organised? He kept accurate data. He could always tell how much of his debt was outstanding by counting the number of strokes left. Those strokes might not mean anything to someone else but they mean a lot for the old man.

Look at the table below and try and make some meaning out of it.

Table 1.1.1: Three Ways of Recording Data

	Column 1	Column 2	Column 3
Sunday	***	!!!!	12
Monday	*****	!!!!! !!!!	3
Tuesday	*****	!!!! !!!	3
Wednesday	*******	!!!!! !	3
Thursday	*****	!!!!!	5
Friday	***	!!!!!	6
Saturday	*****	!!!	7

You may not know the events that were recorded but you can see the frequency of occurrence for each day of the week. Column 1 shows how a little boy recorded the number of cars that came to his father's house during a particular week. Column 2 shows the number of cows a dealer sold in a space of one week. The record was kept by his son. Column 3 illustrates the number of phone calls a young lady received during a particular week. Let look at another table below

Table 1.1.2: Data from a Local Government Election

Ward	**Number of Registered Voters**	**Number of Votes**
1	**1,567**	**1,345**
2	**833**	**823**
3	**1,234**	**1,090**
4	**1,400**	**1,390**
5	**1,556**	**1,460**
6	**1,456**	**1,240**
7	**2,034**	**1,900**
8	**1,902**	**1,890**
9	**1,890**	**1,790**
10	**987**	**970**
11	**456**	**450**
12	**1,340**	**1,322**

The statistical data of the last Local Government election in 12 wards of a Local

government is captured in Table 2 above

Consider the votes on four motions in a state house of assembly. The following data were generated

Table 1.1.3: Data from the Votes on Four Motions in a State House of Assembly

	YES	**NO**	**ABSENT**
Motion 1	18	7	2
Motion 2	11	12	3
Motion 3	13	8	6
Motion 4	9	17	1

2.1.1 Sources of Data

1. Internal Source: When **data** are collected from reports and records of the organization itself, it is known as the internal source.For example, a company publishes its '**Annual Report'** on **Profit and Loss**, **Total Sales, Loans, Wages** etc.

2. External Source: When data are collected from outside the **organization**, it is known as the external source.For example, if a **Tour and Travels** Company obtains **information** on '**Karnataka Tourism**' from Karnataka Transport Corporation, it would be known as external sources of data.

In-text Question

What is Data?

Answer

Data may be regarded as symbols or figures that have potential value or to which meaning can be given.

2.2.0 Data Processing

To put it simply, **data processing** is what we do to **data** in order to make some sense out of them. There are many ways of processing **data**. We may just inspect

data and be able to see a pattern in them. From such a pattern we can make a statement about what has happened and even take a decision. This is visual inspection. Let us apply it to the set of **data** in Table 1. The data in column 1 show that more cars came to the house on Wednesday. Column 2 not only tells us that more cows were sold Monday, but that sales declined during the rest of the week. From column 3, we can say that young lady received the highest number of phone calls on Sunday. We can add that she is more likely to receive more calls during the weekend.

We can build more stories on the **data** in table 2. In each ward, we can compare the number of registered voters with the number of those that voted. Naturally, not everyone who registered could have, voted. We can compute the voting rate for each ward as the percentage of the registered voters who voted. We can find the ward with the highest voting rate and the one that comes next. We can also rank the wards according to the number of registered voters and on the basis of the ranking predict in which wards to expert the highest number of voters. Now, here is a sticky point: what do you say when the number of voters is higher than the number who registered? A big mistake somewhere or what?

Now let us turn to Table 3. Can you say something about the popularity of the four motions? Obviously, motion 1 was the most popular. Motion 2 must have been highly controversial. Motion 4 was highly unpopular.

Besides visual inspection, rearrangement or sorting, **data** processing could take the form of arithmetic processes such as computing the sum or mean. Statistical techniques may also be used to show how much the **data** deviate from some reference value or the relationships within the **data**. Then we are able to make statements about the data as well as generalise our observations to other similar situations. Suppose we found that out of 420 boys that sat for mathematics in SSSE, 303 passed with credit and above; whereas out 392 girls who sat for it the number who passed with credit and above was 121. Then we can at least say that boys do much better than girls in mathematics in that school. If we collected our

data from a number of schools across the country and got the same pattern, then we can say that in Nigeria, boys are better in mathematics than girls provided that the **data** were generated properly.

You may find it difficult to estimate the volume of information a person handles every day. Unless a person is sleeping, his or her brain is always busy processing data and handling information. The brain receives both data and information through the eyes, for instance from what the person reads or sees; through the ears, for instance from what the person hears or listens to; through the nose, for instance from what the person smells, and through the senses of touch, exposure, physical contact and taste.

In essence data processing is simply the conversion of **raw data** into meaningful information through a process. This process follows a cycle which is referred to as **Data Processing Cycle**. **Data** Processing Cycle is a series of steps carried out to extract useful information from **raw data**. These series of steps are in six stages namely: Collection, Preparation, Input, Processing, Output and Storage. Read the six stages of **data** processing at http://medium.com

2.2 Information

In the paragraphs above, we considered the processing of **data**. We found that having done such processing, we were able to make some statements on the situations or events on which the **data** were obtained. Such statements could be used to guide our future response or action.

Fig 1.1.2: Information

This is the function of information: to guide a person on what to do, how and when to do it. Consequently, information may be defined as a fact or set of facts

that can influence a person's response in a given situation. Ideally, information reduces or even eliminates a person's sense of uncertainty. We could also define information as the outcome of **data** processing.

Information is a vital resource. It would be impossible to live a normal life without having adequate information. Just imagine waking up one morning to find that you are alone in the house. Not even one other person is around in your house, in your compound, and in the neighbourhood. A thousand questions are racing through your mind, but there is not a soul around to answer you. You decide to move down some distance but still there is not a single person anywhere. What would you do? For how long would you be able to endure that experience?

In order to embark on tertiary education, you needed information to decide what programme to choose. After completing your application for admission, you waited with some anxiety for, information on whether you have been given admission or not. You did not know what else to do until you got that information.

A businessman would need up- to-date information on the market situation, and changes in government policies that may affect his business. He would normally consult on a regular basis with his colleagues and share information with them.

Today, people are realising more and more the value of information and so a science has developed around it. Information science is the science that deals with the generation, acquisition, organisation, storage, retrieval, dissemination and use of information, its characteristics as well as its impact at the individual, corporate and societal level.

You will agree that information plays a key role in every sphere of life. Information is at the core of success of both individuals and corporate bodies in commercial and business enterprises. Information confers competitive advantage on those who have it against their counterparts who do not have it. Information gives power. The countries that have the capacity to acquire or generate and

manage information use it to improve their socio- economic status and advance ahead of other nations that do not have such capacity.

We that live in the Third World countries, do not seem to fully appreciate the value of information. In many of these countries, most people in the civil service and in government seem to regard information as what government wants the citizenry to hear, the kind of releases made by the minister of ministry of information, or the news one hear from the radio and television or reads in the newspaper. That is information quite alright; but it is "soft" information. The kind of information that confers power in our age, include, scientific, technological, economic and developmental information.

The crucial importance of information has also dictated that both organisations andnational governments take appropriate measures to put in place the infrastructure necessary for managing and possibly for controlling it. More and more investment is being made in the establishment of information systems. National, regional and global computer and telecommunication networks have been developed for the management of communication of information. The countries in the third world have been talking of a new information order. That is the reaction against the domination by the western and industrialised countries of the global information industry

2.2.1 Value of Information

The value of information is enhanced by its accuracy, relevance, timeliness, source, up-to-datedness, as well as the packaging format. If you have any reason to doubt the accuracy of a piece of information, you would not like to act on it. As a matter of fact, inaccurate information could be more disastrous than no information. If it becomes necessary to verify information coming from a particular channel, the cost of such information will be higher, and that may discourage the use of it.

It is quite important that information be relevant to the purpose for which it is

needed. If you would like to read something about the industrial revolution and library staffs give you a book on the colonisation of Africa, would you be pleased? You may accept the book if you think you might have time to read it, but your need for information on the industrial revolution has not yet been satisfied. You may in fact reject the book. The book is still important and will be useful to someone else but not to you at that point in time.

The other consideration is timeliness. For information to be useful, it must be received in good time to make a difference in what the person who receives it is actually doing. If you wanted the information on industrial revolution in the course of preparing for an examination and you could not get it until after that examination, you would no longer attach much importance to it.

The source of information could be very important. The source could have a high level of credibility so that the person using the information could do so with much confidence. Otherwise it could have some qualified level of credibility. Certainly you would be more assured of the authenticity of information when you hear that it has come from an impeccable source.

It must be noted that information has to be presented in a way that makes it easy to use. A brief summary may be enough and much better for a business executive than a voluminous report. Graphic representation may carry more impression than pages of a statistical report. A video presentation could be better appreciated than a written version on a particular subject. The reverse may be the case in another situation.

In-text Question
What is Information?

Answer
Information may be defined as a fact or set of facts that can influence a person's response in a given situation.

3.0 Study Session Summary and Conclusion

This study session has clearly explained the meaning of **data** with relevant examples, how **data** can be processed into useful information and what information entails and its importance. Tutor marked assignment was given to test the students' ability to distinguish between **Data** and Information. Self-Assessment question were provided for students' practice

4.0 Self-Assessment Questions

1. What is data? Illustrate with relevant examples
2. How do you process data to become useful information
3. Name four ways of processing data. Give examples for each
4. Explain the concept of Information
5. Name six factors that determine the value of information and explain
6. With relevant examples explain the six stages of Data Processing

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2C14W1y. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2HNFRZT and critique it in the discussion forum

6.0 References/Further Readings

Susan, A. (1972). *An Introduction to Computers in Information Science*. Metuchen, N.J.: Scarecrow.

Alexander P.M. (2013). Towards reconstructing meaning when text is communicated electronically. Chapter 3 Data, Information and meaning (page 61-72). (http://repository.up.ac.za/dspace)

Study Session 2

Information Retrieval Definitions and Concepts

Section and Subsection Headings:

Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Information and Documents
        2.1.1- Characteristics of Documents
    2.2- Information Retrieval Definitions and Concepts
        2.2.1- Components of Information Retrieval
3.0Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 2, in the previous session, you learnt the basic concepts **data and information**, how data can be processed to make useful information following the **Data Processing Cycle** and the value of information. In as much as the value of information cannot be overemphasized, it is imperative that the way information is stored and retrieved be known therefore this study session will provide you with what information retrieval is all about and the components that made up the retrieval process. However, document will be a reoccurring term in this course so we will need to understand the meaning and its distinction from information.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Distinguish between information and document
2. Describe the characteristics of documents
3. Define Information Retrieval
4. Explain the concept Information Retrieval
5. Understand the different components of Information Retrieval

2.0 Main Content

2.1 Information and Documents

Having discussed in detail what **information** is all about and the distinction from data in study session 1. You will be referred to read more on **information** on study session 1. This section will therefore dwell more on the relationship between information and document. A document is a source of information, and it is as useful as the information in it. A letter from your, uncle is a document. So also is every other letter you receive. Your birthday certificate is a document.

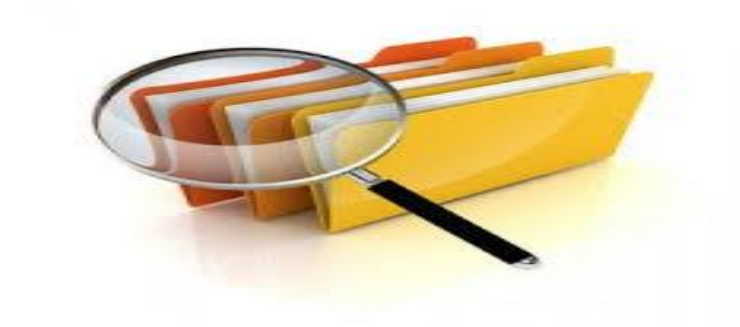


Fig 1.2.1: Information and Documents

It says something about your birth. Every book you buy and add to your **collection is a document**. The minutes of the meetings of your club are documents. Your photo albums and your photographs are more examples of document. Your diary is another document. We can add to the list such documents as an issue of a **newspaper or a magazine, an audio tape with music or speech, a video recording** of any event, a film strip ofa local festival, directories, sales promotion brochure and leaflets, painting and maps. Now you can see that when we use the term "**document**" we are actually referring to a wide range of

information sources. A document is therefore the vehicle conveying information.

2.1.1 Characteristics of Documents

We shall now examine some of the characteristic of a **document**. The first fact to note is that every **document** must originate from somewhere. In many cases, there is one individual or more persons who created the **document**. A person who created a document is the author. If the **document** is a book, the author did all the work of writing it. This **course material** you are reading now was written by somebody who is the author. A book may have one author or multiple authors. Sometimes, a number of authors contribute sections or chapters of a book while one of them or someone else compiles these contributions and edits them to produce the book. The latter is the editor. On the front back of such a book you will find something like

Democracy in Nigeria **Edited by O.O Galu**

Fig 1.2.2:Characteristics of Documents

Sometimes the originator of a **document** may not be just an individual or some individuals but rather an **organisation**, for example the National Universities Commission, the Ministry of **Information**, UNESCO, and so forth. That is a case of co-operative authorship. Even though somebody in the organisation wrote the material, it was written in the name of the organisation. Later in this course you will see the importance of author in locating information.

Another feature, of a **document** is the place of publication. When a **document** has been produced in a large quantity for sale or distribution, it is said to be published. The place where the work of preparing the **document** was done is the place publication. Usually, the place of publication is the place where the publisher has its main office. However, a publisher may publish a **document** in more than one place. Take any book and inspect the title page or the reverse of the title page and

you will find where the book was published. There you will also find the publisher. There are a number of well-known publishers in Nigeria. There are also many little-known publishers.

After the author, the title of the **document** follows in a bibliographic record. The title is the name that the author gives to the **document**. The title could be in two parts: the main title followed by a sub-title, for example: "**Computer Concepts: A User Perspective**." Besides identifying a document, the title could also give a glimpse into what the **document** is about. The title above suggests that the book is about computers. It must be noted that this is not always the case. If you see a document with the title, "**An Enterprise in Futility**" would you be able to guess what it is all about?

Another feature that may identify a **document** is the edition. A new edition comes out when a **document** is revised. Revision is usually necessary for the purpose of correcting errors in a **document**, enhancing aspects of the **document**, supplying new information, expanding the scope of coverage and so forth. Some documents go through many revisions, which generate successive editions. Two different editions of the same **document** are actually regarded as two different documents.

The date of publication is no less important. The date of publication says how recent the **document** is. In some subject field publications become quickly out of date.

Lastly, we should mention the unique identification number that every book should have. It is called the ISBN (International Standard Book Number). It is a 13 digit number, for instance 978-1-846-14792-0. ISBN not only identifies the book, but also the publisher and the country of publication. Serial publications such as journals and magazines also have a unique identification number called ISSN (**International Standard Serial Number**).

You may wonder if every **document** has all the features described above. Not

every document has them. A **document** that has not been published cannot have a publisher or place and date of publication. We may classify information source as published or unpublished. The value of an information source does not depend as much on whether or not it has been published as to the novelty and potential usefulness of the information it contains

In-text Question
What is a document?

Answer
A document is a source of information, and it is as useful as the information in it.

2.2Information Retrieval Definitions and Concepts

Information Retrieval has been defined in several ways which in turn have the same meaning and interpretations. Here are several definitions of Information Retrieval

1. A process of searching for specific **information** among a large collection
2. It is a discipline concerned with the process by which queries presented to **information retrieval systems** are matched against a store of documents
3. It is a process in which sets of records or documents are searched using queries or terms to find items which may help to satisfy the information need or interest of an individual or a group

All the above three definitions are acceptable for **Information Retrieval**; however, the last definition can be taken as more appropriate than the other two because it is more loaded in terms of the components that constitute the process of **information retrieval** which will be discussed in details in the next section.
Information could be in electronic or paper-based form which means **information** can be retrieved both manually and electronically. However, this course is more of the electronic retrieval rather than manual retrieval. For any of these (**manual or electronic**), the way a user express their **information needs** help in **information**

retrieval. It is worth mentioning that **information retrieval** can be through an intermediary who could be human or computer. This is why **information retrieval** is also a discipline that people study.

In a computerized **information system**, as shown in figure 1below, we have the human with an information need interacting with an information retrieval system (**by query using search terms**) through an interface (**normally a hardware containing software with many options**) to produce information items(**in the document store or database**) that matches the information need of the human. Figure 1 shows this illustration

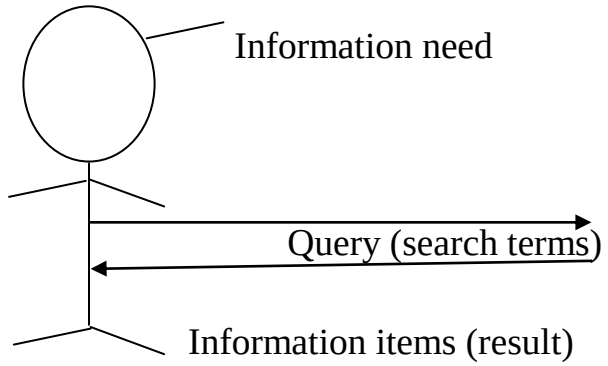


Figure 1.2.3: Illustration explaining Information Retrieval

Document store/ database matching user query with items in store

Interface (a hardware containing a software with many options

2.2.1 Components of Information Retrieval

As mentioned earlier in the preceding section, **information retrieval** process consists of some components which include the following:

1. The set of record or documents that needs to be searched

   For any information to be retrieved, this means that **information** should be somewhere hidden waiting to be found. This is where the sets of records or store of documents comes in. Examples of these document stores are databases in computerized systems, documents on any search engine like Yahoo, Google, Google Scholar etc. **Institutional Repositories**, Directories of Open Access Journals or Books where electronic journals and books are stored, specialized databases such as HINARY, JSTOR, Science Direct and so on.

2. The indexing and access method for the document set

   Every document set has a way it is indexed and organised for easy access. Depending on the indexing and access method for the document set a user can retrieve **information matching** his/her **information need**(s). Examples are using Boolean method, Truncation or Natural language for searching and so on.

3. The **information need** of the user

   Every **information searcher** has a need. This need could be trivial or scholarly, either of this two requires you go for searching so as to be able to satisfy your information need. If there is no **information need** then there can be no **information retrieval**.

4. Search Strategy

   It is very important to be able to verbalize or express one's information need in a series of search statement (**search strategy**). In order to retrieve information that matches a user's **information need**, the need must be broken down in form of search statement which is known as Search Strategy. Search strategy is a technique of carrying out a search in an **information retrieval** system. Expressing one's need is essential in **information retrieval** as this can determine the quantity, quality and relevancy of information retrieval products.

5. Search Results

The sequence of **information** items presented as a result of the search carried out by a user that has an **information need** using a search strategy is known as search

results. In as much as a user with an **information need** as posed a query in form of search statement to an information retrieval system, he/she is expecting a result in form of information item that will match his/her information need

The diagram below is a good illustration of the components of **Information Retrieval** process

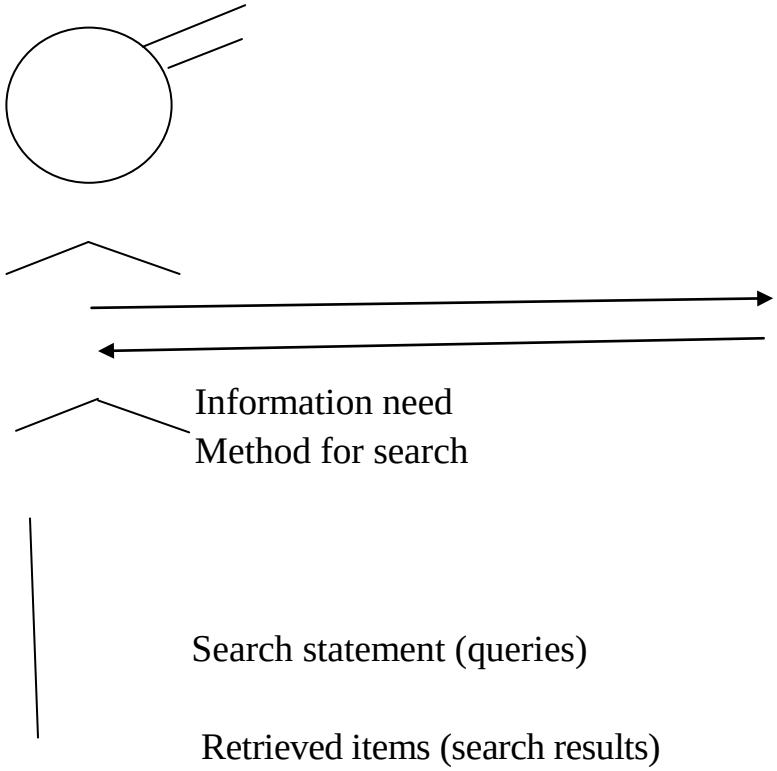


Figure 1.2.4: Components of Information Retrieval

Set of records or Document store

In-text Question

Define Information Retrieval

Answer
Information Retrieval is a process in which sets of records or documents are searched using queries or terms to find items which may help to satisfy the information need or interest of an individual or a group

3.0 Study Session Summary and Conclusion

This study session has clearly distinguished between **information and document**. Also the characteristics of documents and its importance in **information retrieval** were discussed. Several definitions and components of information retrieval were enumerated and explained in details with diagrammatic illustrations. Tutor marked assignment was given to test the students' ability to distinguish between **Information and documents**. Self-Assessment question were provided for students' practice

4.0 Self-Assessment Questions

1. Distinguish between information and documents
2. Clearly explain the term "Information Retrieval"
3. With relevant examples explain each of the components of a retrieval process

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2NGfUOk. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/30DPN0C and critique it in the discussion forum

6.0 References/Further Readings

Ricardo, B. and Berthier, R. (2015). Modern Information Retrieval. ACM Press New York. https://www.researchgate.net/publication/

Study Session 3

Information Retrieval System

Section and Subsection Headings:

Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Information Retrieval System
        2.1.1- Types of Information Retrieval System
    2.2- Data/fact Retrieval System versus Information Retrieval System
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 3, in the previous session, you learnt the distinction between **information and documents**, also the concepts of Information retrieval and components were introduced to you. This session will however explained the meaning of **Information Retrieval System** and their types. The differences between Data/fact **Retrieval Systems and Information Retrieval System** were also enumerated.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain the concept Information Retrieval System
2. Identify and explain the types Information Retrieval System
3. Identify the distinguishing factors of DataRetrieval and Information Retrieval Systems

2.0 Main Content

2.1 Information Retrieval System

An **Information Retrieval** (IR) system is a manual or computerized process for storing, organising and accessing **information items**. It can also be defined as any device which aids access to document specified by subjects and the operations associated with it (Vickery, n.d)). Hiemstra (2000) defined IR system as a software programme that stores and manages **information on documents**. The documents can be books, journals, atlases and other records of thought, or any parts of such records- articles, chapters, sections, tables, diagrams or even particular words. The operations can range from simple visual scanning to the most detailed programming.

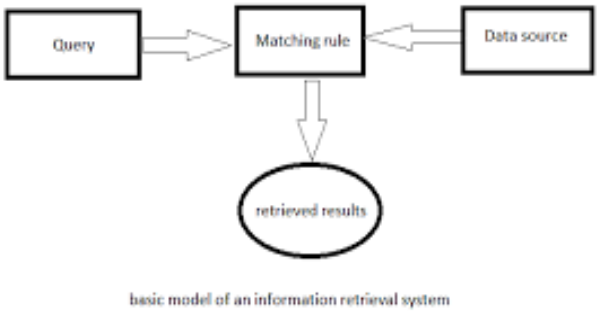


Fig 1.3.1: Information Retrieval System

An IR system is designed to retrieve any documents or **information** required by a user. An IR system stores units of **information**. Each unit is linked in the system to specifications of one or more documents or part of documents. The user specifies particular unit of **information and the system** is designed to provide him with a knowledge of a relevant items recorded in the system. This implies that IR system does change the knowledge of the user on the subject of his enquiry; it merely informs the user of the existence or non-existence and whereabouts of documents relating to the request(Aruleba, AkomolafeandAfeni, 2016).

The act of **retrieving information** is possible for only information that has been stored in the system. The problem of **information retrieval** is basically of selecting from a collection of document which is/are expected to be useful to the user. This is not much of a problem if the collection is small, a user can browse through the

collection, examine each item and determine which one he want. The problem arise if the size of the information is so large that it must be organised for searching

The main issues of **information retrieval** is first, the organisation of the collection and second, retrieval. A searcher or requester must be able to describe the attributes of the document in which he wishes to inspect and the system in turn must be able to identify the and search the collection based on the attributes, summarily, an IR system does not inform the user of the subject of its enquiry. It only informs him of its existence and where about of the document relating to his request.

In an IR system, because the format and language used in storing document are less constrained, the IR system can retrieve document or **document references** and can store a body of **information** which can be searched and retrieved in response to specific request or needs. The ability of the system to do this depends to a large extent on how the **documents** have been stored. The indexing of document through the use of control indexing terms allows some form of structure to be imposed on the **organization of document** in the system and this is convenient tool when searching is to be carried out.

A request put to an IR system that uses a control vocabulary is handled in the same way as an incoming document. The request is conceptually analysed and described by means of terms selected by the controlled vocabulary. The request in controlled language is now searched against the subject index and a document is retrieved when a match or partial match occurs between a formalized concise description of the document and the request. Searching is thus essentially a matching operation and the subject indexing or control vocabulary is a tool that reduces a number of document a searcher needs to look at in order to find one or more documents that will satisfy his/her information needs.

The first/initial output from the search process may be in form of citation from which the requester can make a selection. Subsequently, the requester can ask for

the complete text of the selected item to be presented if the system offered such facilities. IR system can be categorised into two types as discussed below

2.1.1 Types of Information Retrieval System

There are usually two types of IR system. They are Bibliographic Retrieval System and Full Text Retrieval System

Bibliographic Retrieval System: An IR system may retrieve complete text of document surrogate such as **abstract or bibliographic citation**. This can also be referred to as Reference **Retrieval System**. This type of IR system normally contains references to the original document and contains things like title, abstract, publishers, authors and sometimes introduction and so on. They can be entered into the system and they are also searchable. It sometimes contains index terms that are assigned by **human index**. Although other contents such as title, abstract etc are also entered into the system and are usually searchable. For example, a user (you) might request for all **bibliographic reference** dealing with "Information System Analysis and Design". The system will only bring out the reference to the request like authors name (s) year of publication, title etc. that can refer the user to where the full text of the request can be found.

Full Text Retrieval System: This type of IR system on the other hand, contains the complete text of the **document and the database** is usually indexed based on the words present in the entire document.

A system that ultimately provides the user or searcher with full text of document is properly called Document Retrieval System. The task of a full-text **information retrieval (IR) system** is to satisfy a user's **information** need by identifying the documents in a collection of documents that contain the desire information .(Brown, Callan, Croft, and Moss, 1993).In Full Text RS, the search is distinguished from searches based on parts of the original texts represented in databases (**such as titles, abstracts, selected sections, or bibliographical references**).

In a full-text search, a search engine examines all of the words in every stored document as it tries to match search criteria (**for example, text specified by a user**). Many websites and application programs (such as word processing software) provide full-text-search capabilities. Some web search engines, such as AltaVista, employ full-text-search techniques, while others index only a portion of the web pages examined by their indexing systems.

In-text Question
What is Information Retrieval System?

Answer
An Information Retrieval (IR) system is a manual or computerized process for storing, organizing and accessing information items.

2.2 Matching Document and Query

Suppose there is a store of **documents** and a person formulates a question (request or query) to which the answer is set of **documents** satisfying the information need expressed by his question. He can obtain the set by reading all the documents and discarding all the others. In a sense, this constitutes "Perfect Retrieval".

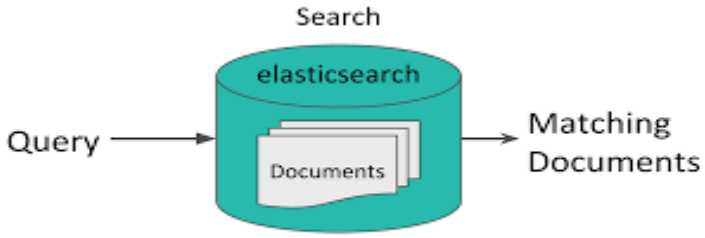


Fig 1.3.2: Matching Document and Query

However, an IRS is built on the concept of Query (q) and **Document** (d). An IR system must match q and d or d and q depending on the user interface or the direction of the system orientation. Matching is what an IR system does. In relation to defining what an IR system does, it should be noted that IR system matches user queries to documents stored in a database.

Since matching of q and d and d and q is the function of an IR system, all

components of IR system must support its matching ability. Thus, an IR system must perform the following tasks of:

1. Translating an information need into query terms (q)
2. Indexing documents into index terms (d)
3. Matching the query and documents
4. Ranking the results of matching
5. Managing the access to documents

2.3 Data/fact versus Information Retrieval System

There is a difference between **Data Retrieval** (DR)system and Information Retrieval (IR) system the major difference between the two is that a DR retrieves the actual data desired i.e, there must be an exact match between queries and the item retrieved e.g, today's date remains today's date, it cannot be changed likewise the shared price of Nigerian breweries as of today remains the same. The capital city of Nigeria as at today remains the same and so on. For IR, system they provide a set of **information** that will most likely contain the information the user wants.

The user now has to check the retrieved items to see if it matches his needs. (It is subjective). There is always an element of uncertainty because of a continuum of irrelevant to relevant **information document** search. Following the preceding section, a distinction is needed since the two terms affect the matching type: Exact match and best match. This will be illustrated in the table below

Table 1.3.1: distinction between Data Retrieval (DR) and Information Retrieval (IR) systems

	Data Retrieval (DR) System	**Information Retrieval (IR) System**
1.	Deals with exact match i.e. whether an item is present or absent in the system	It is more concerned with partial and best match i.e. close to what the user desire
2.	Probabilistic is not considered in DR. it only focuses on matching of items and thus is sensitive to small errors	Focuses on relevance and thus is not significantly affected by small errors
3.	Determines which document contains the keywords given by the users therefore may	Retrieves information about a subject

	not meet the information need of the user	
4.	A user of DR system does not tolerate a single erroneous object among a thousand retrieved objects and perceives it as a total failure of the system	Small errors could go unnoticed by the user of IR system
5.	Handles data with a well defined structure and semantics	Handles natural language texts-unstructured and ambiguous in semantics
6.	Refers to the physical description of an item that calls for objective indexing like the name of an author	Refers to information contained or meant in an item. Calls for subjective indexing
7.	In respect to the evaluation of retrieval product based on performance (precision and recall). DR is expected to have a performance100% recall and 100% precision	IR could have a performance of 30% recall and 30% precision

Note Item 7 On The Table

***1.3.2:* Recall indicates the ability of the system to retrieve relevant items, while precision is the ability of the system to reject irrelevant items.**

In-text Question

What are the functions of an IR system?

Answer

1. Translating an information need into query terms (q)
2. Indexing documents into index terms (d)
3. Matching the query and documents
4. Ranking the results of matching
5. Managing the access to documents

3.0 Study Session Summary and Conclusion

This study session has clearly explained the **Information Retrieval System** and the types (**bibliographic and full text**). This session also distinguished between **Data Retrieval system** and **Information Retrieval system**. Tutor marked assignment was given to test the students'. Self-Assessment question were provided for students' practice.

4.0 Self-Assessment Questions

1. What is Information Retrieval System?
2. Clearly explain Bibliographic and Full-text retrieval system.
3. Distinguish between Data/fact Retrieval System and Information Retrieval System

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/1bhYPOq. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/30DPN0C and critique it in the discussion forum

6.0 References/Further Readings

Baeza-Yates, R. and Ribeiro-Neto, B. (1999).*Modern Information Retrieval*. Addison-Wesley

Chowdhury, G.G (2003). *Introduction to Modern Information Retrieval*.Neal-Schuman

Modules 2

Information Retrieval Models

Contents:

Study Session 1: Information Retrieval Models: Introduction
Study Session 2: Boolean Model and Extended Boolean Model
Study Session 3: Vector Space Model and Fuzzy Model
Study Session 4: Probabilistic and Natural Language Model

Study Session 1

Information Retrieval Models: Introduction

Section and Subsection Headings:

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1.0 Learning Outcomes
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    2.1- Information Retrieval Models at a glance
    2.2- A Conceptual Model of Information Retrieval
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to module 2 of this course, in the previous session, you were introduced to **Information Retrieval Systems**, the types and the basic distinction

between **Data Retrieval System and Information Retrieval System**.
This session explains the concept "**Information Retrieval Models**".

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain Information Retrieval Models
2. Explain the concept Information Retrieval Models

2.0 Main Content

2.1 Information Retrieval Models at a glance

Your primary goal of **information retrieval** (IR), is to provide users (you) with those documents (**including non-textual information, such as multimedia objects**) that will satisfy their information need. You have to formulate your **information need** in a form that can be understood by the system.

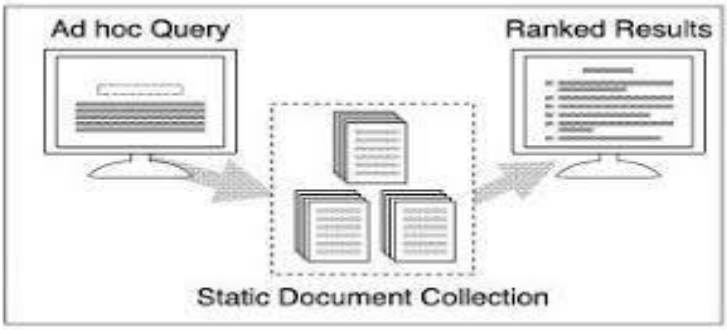

Fig 2.1.1: Information Retrieval Models at a glance

Information seeking is a form of problem solving. It proceeds according to the interaction among eight sub processes:

1. Problem recognition and acceptance
2. Problem definition
3. Search system selection
4. Query formulation
5. Query execution
6. Examination of results (including relevance feedback)
7. Information extraction and

8. Reflection/iteration/termination.

To be able to perform effective searches, you have to develop the following expertise:

1. Knowledge about various sources of **information**,
2. Skills in defining search problems and applying search strategies,
3. Competence in using electronic search tools.

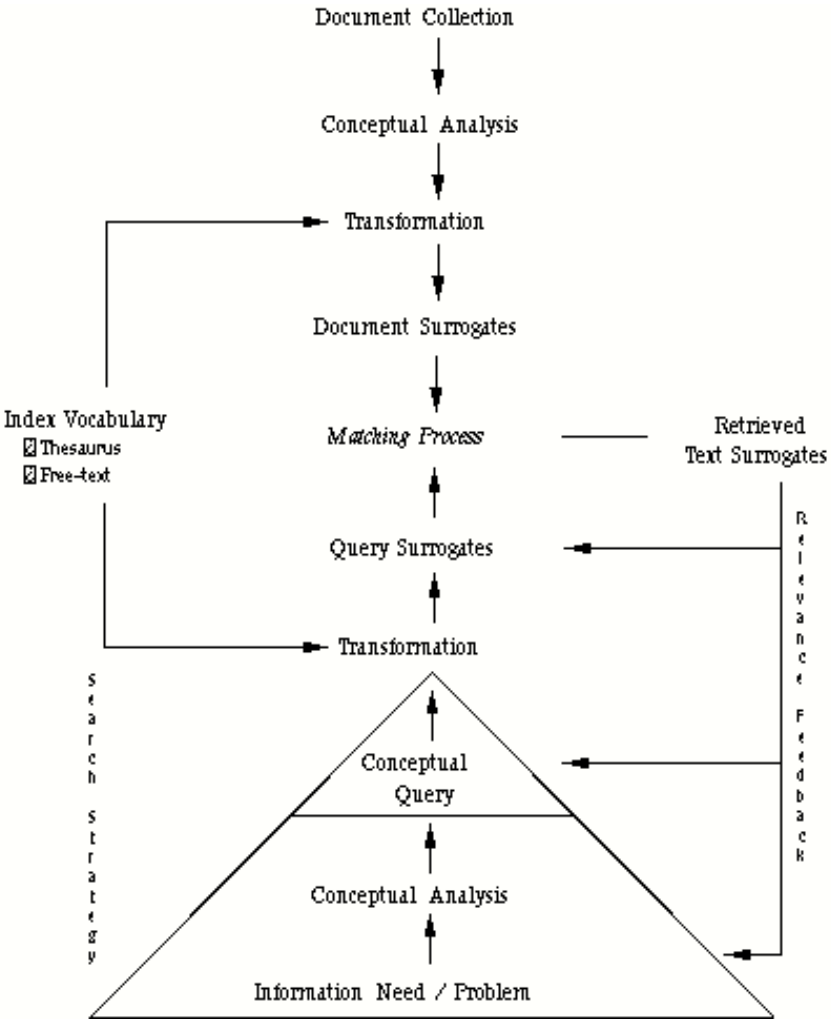

Figure 2.1.1: Representation of General Model of Information Retrieval Process

Information Retrieval Process The **information need** of the user can be understood as forming a pyramid, where only its peak is made visible by users in

the form of a conceptual query (figure 3). The conceptual query captures the key concepts and the relationships among them. It is the result of a conceptual analysis that operates on the **information need**, which may be well or vaguely defined in the user's mind. This analysis can be challenging, because you are faced with the general "**vocabulary problem**" as they are trying to translate their **information need** into a conceptual query.

The vocabulary problem refers to the fact that a single word can have more than one meaning, and, conversely, the same concept can be described by surprisingly many different words. It has been shown that two people use the same main word to describe an object only 10 to 20% of the time. Further, the concepts used to represent the documents can be different from the concepts used by the user.

The conceptual query can take the form of a natural language statement, a list of concepts that can have degrees of importance assigned to them, or it can be a statement that coordinates the concepts using Boolean operators. Finally, the conceptual query has to be translated into a query surrogate that can be understood by the retrieval system.

In-text Question
What is the goal of information retrieval?

Answer
The goal Information retrieval (IR) is to provide users with those documents (including non-textual information, such as multimedia objects) that will satisfy their information need.

2.2 A Conceptual Model of Information Retrieval

When you are dealing with text or multimedia documents, you should distinguish different views on these **documents.** Several subfields of computer science and related fields deal with documents, where most of the fields focus on one or two views and ignore the others. We will try to present an integration of the different perspectives. Four different views are distinguished below: http://www.is.informatik.uni-duisburg.de/bib/pdf

External attributes: comprise data that is not contained within the **document**, i.e. a user (you) looking at the document only may not see these values. External attributes contain **information** that is needed for certain types of processing the document, e.g. the name of the creator of the document, access rights, or publication **information**. In digital libraries, this type of data often is called metadata.

Logical structure: The media data that is contained within the **document** and its internal structure comprise the logical structure of a document. Usually, **documents** have a hierarchical structure (e.g. a book divided into chapters, chapters containing sections, consisting of subsections, which comprise paragraphs, images and tables). In this tree structure, the data is located in the leaves, where a leaf contains single media data only (e.g. text, graphics, images, audio, video, animation, 3D). Hypermedia links allow for non-hierarchical structures.

Layout structure: In order to show a document to a user, it must be presented at some kind of output media (e.g. **when a document is printed, we have a sequence of pages**). Based on a so-called style sheet, the layout process maps the logical structure onto the output media. The layout structure describes the spatial distribution of the data over the output media, e.g. the sequence of pages, which in turn are subdivided into rectangular areas (e.g. **page header, footer, columns**). This concept can be extended to time-dependent media (e.g. audio, video), where the layout structure describes the temporal and spatial distribution on an appropriate output device.

Content deals with the meaning of a document (e.g.: **What is the document about? What does it deal with**?). The content is derived from the logical structure, in most cases by an automatic process. The content representation may have an internal structure, too, but often rather simple schemes are used. For instance, in text retrieval, content mostly is represented as a set of concepts.

When you want to perform **information retrieval** on multimedia documents, you have to consider all these views, in order to allow for queries addressing each of these views separately, as well as queries for combinations. Examples of queries with respect to the different views are: Give me all documents published last month (**attributes**).
Show me all books that have the string `XML' in the title and contain more than 10 chapters (**logical structure**). Show me all articles that are typeset in two columns, with a length of more than 15 pages (layout). Find all documents about image retrieval (**content**). Since IR focuses on content, you also will prefer this view throughout this paper. However, since real applications Typically involve more than one view, there is a need for retrieval mechanisms that are not restricted to a single view.

A variety of models have been proposed to make retrieval process more efficient.

A model is an abstracts away from the real world, which uses a branch of mathematics and possibly a metaphor for searching. Each retrieval model is unique in itself and a classification based on only one component regardless of how major it could be, hides its true potentials. IR models are few and upon limitations, extended and improved. Implementation of models varies even for similar data type and this is very critical issue in commercialized IR systems.
The well-known **information retrieval** models include:

1. Boolean model
2. Extended Boolean Model
3. Document similarity
4. Probabilistic indexing
5. Vector space model
6. Probabilistic retrieval
7. Fuzzy set models
8. Inference networks
9. Natural Language models

However, Boolean Model and Extended Boolean Model, Vector Space model and Fuzzy Set models, Probabilistic Model and Natural Language Model will be discussed in the study session 2, 3and 4 respectively of this module.

In-text Question
What are the four different views that distinguished about the Model of information retrieval?

Answer
External attributes
Logical structure
Layout structure
Content

3.0 Study Session Summary and Conclusion

This study session has made the concept of **Information Retrieval** Modeleasy to understand using the diagrammatic representation of general Model of **Information Retrieval Process**. Different views regarding the conceptual model of **Information Retrieval** were also discussed. Tutor marked assignment was given to test the students'. Self-Assessment question were provided for students' practice.

4.0 Self-Assessment Questions

1. Explain the Information Retrieval Model?
2. Enumerate and clearly explain the different views on the conceptual model of Information Retrieval
3. With the aid of a diagram, explain the general model of Information Retrieval Process

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://bit.ly/2Uk2WZo/. Watch the video & summarise in 1 paragraph
b. View the animation on add/site https://bit.ly/2Wk8WFx and critique it in the discussion forum

6.0 References/Further Readings

Baeza-Yates, R. and Ribeiro-Neto, B. (1999).*Modern Information Retrieval*.Addison-Wesley

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Study Session 2

Boolean Model and Extended Boolean Model

Section and Subsection Headings:

Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Boolean Model
        2.1.1- Advantages and Disadvantages of Boolean Model
    2.2- Extended Boolean Model
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 2, in the previous session, you were made to understand the **Information Retrieval Model** as a concept. This session and the subsequent ones in this module will be discussing the different types of IR **Models**. We will discuss the **Boolean Model** and the Extended **Boolean Model**, their advantages and disadvantages.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain Boolean Models with their illustrations
2. Explain Extended Boolean Models with their illustrations
3. Enumerate the advantages and disadvantage of both Models

2.0 Main Content

2.1 Boolean Model (BM)

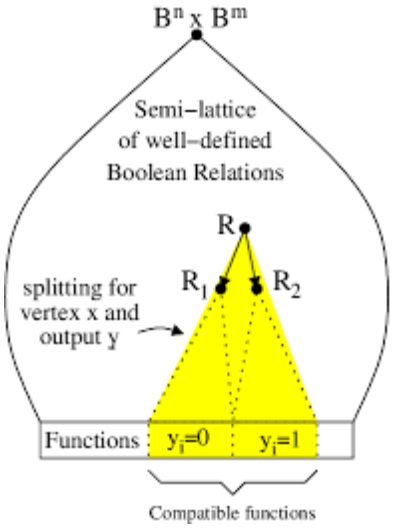

Fig 2.2.1: Boolean Model (BM)

From our discussion, The **Boolean model** of **Information Retrieval** is a classical Information Retrieval (IR) model and, at the same time, the first and most adopted one. Proposed in 1950by a mathematician named George Boole. BM is used by virtually all commercial information retrieval systems today.

The **Boolean information retrieval** is based on Boolean Logicand classical Set Theory in that both the documents to be searched and the user's query are conceived as sets of terms. The model uses Union, Intersection and Negation operators (**OR, AND, NOT**) retrieval is based on matching procedures where items retrieved in response to query are either relevant or irrelevant. This is due to the extreme character of the conjunction and disjunction. Retrieval is based on whether or not the documents contain the query terms.

Boolean operators are devised to enhance recall and precision. Recall reflects the ability to retrieve relevant documents and precision reflects the ability to reject irrelevant document. AND is intended for precision while OR is intended for recall. BM is evaluated and tested using recall and precision against a binary scale of relevance. Boolean AND is applied between concepts while OR is for terms with a concept.

NOT is used in a cautious manner to avoid missing of relevant items and reserved for searches that strictly require exclusion of terms. For example, yoursearch on images and NOT any other components of multimedia like video and audio.

Examples:

Boolean AND (Intersection) - all of the words: The **Boolean** AND of two logical statements x and y means that both x AND y must be satisfied.

In order to retrieve the document both terms must be included in the search term. Also the two terms must be in the items retrieved.

Let us look at the diagram below, using the terms "**Information**" and "**Retrieval**" for searching.

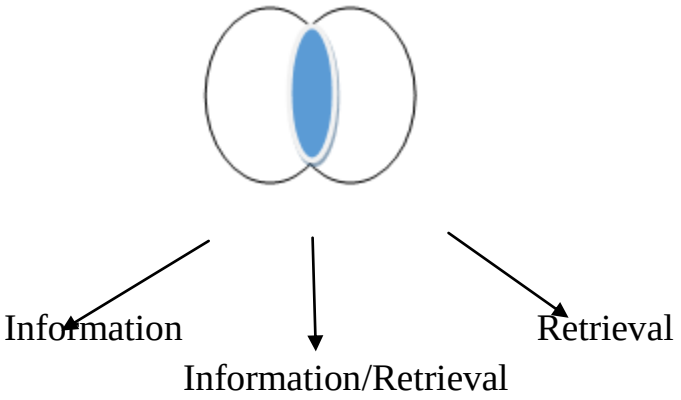

Figure 2.2.2: Diagram representing Information AND Retrieval

All documents in the IR system containing **Information/ Retrieval** (joined in a document) are retrieved as **information** items. That is, the shaded part infigure3

Boolean OR (Union)– any of the words: **Boolean** OR of these same two statements means that at least one of these statements must be satisfied. Document can be retrieved with any or both the present terms. Information, retrieval and **information** retrieval

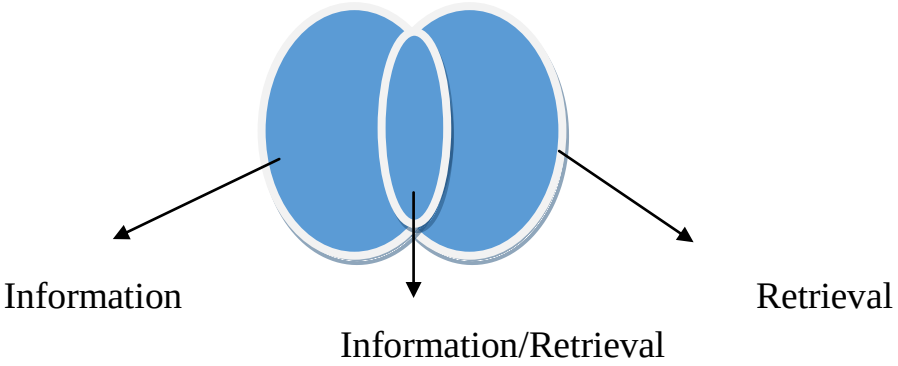

Figure 2.2.3: Diagram representing Information OR Retrieval

The IR system looks for **information** items containing the terms Information, **Information/Retrieval** and Retrieval. That is, all shaded parts in figure 4

Boolean NOT (Negation) – none of the words: excludes the term that is not needed. As earlier mentioned, this operator should be used in a cautious manner in order not to miss relevant items.

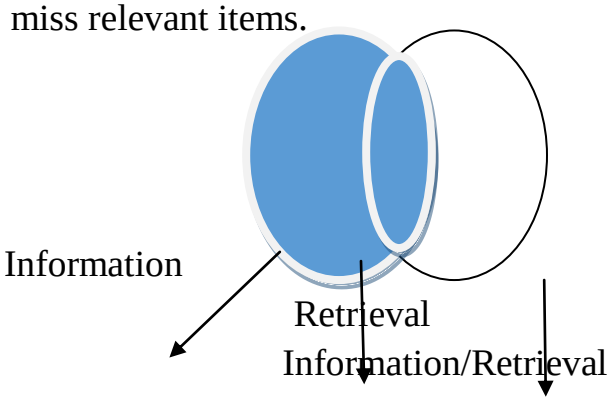

Figure 2.2.4: Diagram representing Retrieval NOT Information

The IR system looks for **information** items containing only "Retrieval". **Information** items that contains any terms with "**Information**" are not included in the search. That is, the un-shaded part of figure 5.

In-text Question
Define Booleanmodel

Answer
Booleanmodelof Information Retrieval is a classical Information Retrieval (IR) model and, at the same time, the first and most adopted one.

2.1.1 Advantages and Disadvantages of Boolean Model

There are several advantages and disadvantages of **Boolean Models** you should have in mind. These are enumerated below:

Advantages of Boolean Model

1. It is easy to implement and it is computationally efficient. Hence, it is the

standard model for the current large-scale, operational retrieval systems and many of the major on-line information services use it.

2. It enables users to express structural and conceptual constraints to describe important linguistic features. Users find that synonym specifications (reflected by OR-clauses) and phrases (represented by proximity relations) are useful in the formulation of queries.
3. Boolean approach possesses a great expressive power and clarity. Boolean retrieval is very effective if a query requires an exhaustive and unambiguous selection.
4. The Boolean method offers a multitude of techniques to broaden or narrow a query.
5. The Boolean approach can be especially effective in the later stages of the search process, because of the clarity and exactness with which relationships between concepts can be represented.

Disadvantages of Boolean Model

1. The major problem with BM is that it is very rigid. They look for the exact word used in the search rather than variant word. Many researchers refer to it a fact or data retrieval system rather than information retrieval system.
2. It gives “all or nothing” response. Too many or few documents are retrieved so the size of the results is difficult to control.
3. It does not rank the output of the query or searches. i.e documents are not ranked in any order of presumed importance to the user.
4. Another disadvantage of Boolean model includes the fact that users find it difficult to construct effective Boolean queries for several reasons. Users use the natural language terms AND, OR or NOT that have a different meaning when used in a query. Thus, they will make errors when they
5. form a Boolean query, because they resort to their knowledge of English.

In-text Question

What are the advantages of Boolean Model?

Answer

1. It is easy to implement and it is computationally efficient. Hence, it is the standard model for the current large-scale, operational retrieval systems and many of the major on-line information services use it.

2. It enables users to express structural and conceptual constraints to describe important linguistic features. Users find that synonym specifications (reflected by OR-clauses) and phrases (represented by proximity relations) are useful in the formulation of queries.

3. Boolean approach possesses a great expressive power and clarity. Boolean retrieval is very effective if a query requires an exhaustive and unambiguous selection.

4. The Boolean method offers a multitude of techniques to broaden or narrow a query.

5. The Boolean approach can be especially effective in the later stages of the search process, because of the clarity and exactness with which relationships between concepts can be represented.

2.2 Extended Boolean Model

To overcome the rigidity of **Boolean Model**, the Extended **Boolean Model** (EBM) was developed by Edward Alan Fox in 1983. In the EBM, the information are judged as not only relevant or irrelevant with respect to a given query but given relative query. The EBM also provide proximity and adjacency operators to compare a query with document index terms.

The Boolean model doesn't consider term weights in queries, and the result set of a Boolean query is often either too small or too big. The idea of the extended model is to make use of partial matching and term weights as in the vector space model. It combines the characteristics of the Vector Space Model with the properties of Boolean algebra and ranks the similarity between queries and documents. This way a document may be somewhat relevant if it matches some of the queried terms and will be returned as a result, whereas in the Standard **Boolean model** it wasn't.

Thus, the EBM can be considered as a generalisation of both the **Boolean** and vector space models; those two are special cases if suitable settings and definitions are employed. Further, research has shown effectiveness improves relative to that for Boolean query processing. Other research has shown that relevance feedback and query expansion can be integrated with extended Boolean query processing.

Extended Boolean Model provides also left and right truncation e.g, searching for Computer Architecture. It will search for everything that begins with Comp# and Arch#. Therefore, for Comp#, it may retrieve computer, computing, computation etc. the drawback of this type of model is that there is problem of false drop e.g. **retrieving information** on computer, to avoid a lot of false drops you can use computer for searching. Another example is #Architecture which is anything architecture. This can be referred to as Root or Left truncation while the former e.g. computer can be referred to as Stem or Right truncation.

The drawback of this **model** is that there is a problem of false drop e.g. in retrieving a word like computer, and you use comp#, you may have false drops such company, composition, compare and so on. In order to avoid a lot of false drops you can use computer for searching

In-text Question
What is the full mean of EBM?

Answer
Extended Boolean Model

3.0 Study Session Summary and Conclusion

Two types of **Information Retrieval Model** have been explained in this study session. They are **Boolean Model** and **Extended Boolean Model**. A diagrammatic representation of **Boolean** operators was presented for easy understanding. Tutor marked assignment was given to test the students on their understanding of both **models**. Self-Assessment question were provided for

students' practice.

4.0 Self-Assessment Questions

1. Write short note on Boolean Model of information retrieval. Explain using diagrams. Give their advantages and disadvantages
2. Write short note on Extended Boolean Model of information retrieval. Give a detailed explanation using example. Give their advantages and disadvantages

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/1hxT0ot. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2YJyIk8 and critique it in the discussion forum

6.0 References/Further Readings

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Study Session 3

Vector Space Model and Fuzzy Model

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Vector Space Model
        2.1.1- Advantages and disadvantages of Vector Space Model
    2.2- Fuzzy Model
3.0Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 3, your previous session, started with the discussion of two types of the **information retrieval models- Boolean and Extended Boolean Model**. This studying session will discuss another two of the IR models which are Vector Space and Fuzzy Models.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain Vector Space Model
2. Explain Fuzzy Model

2.0 Main Content

2.1 Vector Space Model

The **Vector space model** was developed to overcome the rigidity of **Boolean model**. Proposed in 1970, it is a statistical **retrieval model** that represents the documents and queries as vectors in a multidimensional space, whose dimensions are the terms used to build an index to represent the documents. The creation of an index involves lexical scanning to identify the significant terms, where morphological analysis reduces different word forms to common stems, and the occurrence of those stems is computed.

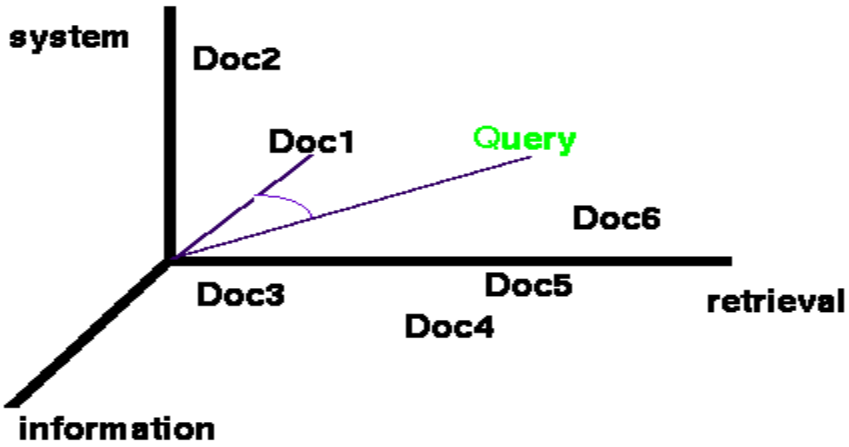


Fig 2.3.1: Vector Space Model

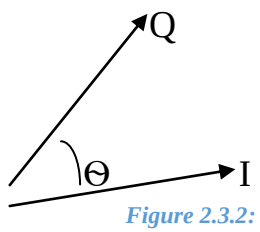
Information items as well as queries are described as binary **vectors** whose components indicate the absence or presence of the nth **indexing term**. Query and **information** items are points in the **vector space** and information items are ranked by the number of common descriptors in items and query. Mathematically, this is the inner product between the two **vectors**.

Query and document surrogates are compared by comparing their **vectors**, using, for example, the cosine similarity measure. In this **model**, the terms of a query surrogate can be weighted to take into account their importance, and they are computed by using the statistical distributions of the terms in the collection and in the documents.

The **vector space model** can assign a high ranking score to a document that

contains only a few of the query terms if these terms occur infrequently in the collection but frequently in the document.

In order to explain the above paragraph more clearly, is the well-knownsimilarity measure which is known as COSINE measure. This is represented by the cosine of the angle between the query and the item vector. The cosine measure ranks the item according to their angle to the query vector. Vectors pointing in similar directions are thus taken to represent similar concepts. A process of clustering is the used to build up groups of items that are close to each other and can be represented by one most typical item.



Q - Query term
I- Information items
Θ- Cosine measure

Figure 2.3.2: Cosine of the angle between the query and the item vector

2.1.1 Advantages and disadvantages of Vector Space Model

The vector space model, like all statistical **retrieval models** have the following advantages:

1. They provide users with a relevance ranking of the retrieved documents. Hence, they enable users to control the output by setting a relevance threshold or by specifying a certain number of documents to display.
2. Queries can be easier to formulate because users do not have to learn a query language and can use natural language.
3. The uncertainty inherent in the choice of query concepts can be represented.

However, the vector space model as well as other statistical models has the following shortcomings:

1. They have a limited expressive power. For example, the NOT operation cannot be represented because only positive weights are used. Again, the very common and important Boolean query ((A and B) or (C and D)) cannot be represented by a vector space query. Hence the statistical

approaches do not have the expressive power of the Boolean approach.

2. The statistical approach lacks the structure to express important linguistic features such as phrases. Proximity constraints are also difficult to express, a feature that is of great use for experienced searchers.
3. The computation of the relevance scores can be computationally expensive.
4. A ranked linear list provides users with a limited view of the information space and it does not directly suggest how to modify a query if the need arises.
5. The queries have to contain a large number of words to improve the retrieval performance. As is the case for the Boolean approach, users are faced with the problem of having to choose the appropriate words that are also used in the relevant documents.

The vector space model, on its own makes the following assumptions:

1. The more similar a document vector is to a query vector, the more likely it is that the document is relevant to that query.
2. The words used to define the dimensions of the space are orthogonal or independent. While it is a reasonable first approximation, the assumption that words are pair wise independent is not realistic.

2.2 Fuzzy Model

The **Boolean model** had imposed a binary criterion for deciding relevance. Thus, the question of how to extend the **Boolean model** to accommodate partial matching and a ranking has been a problem for some time now.

This question was answered by the introduction of the **fuzzy set model** in 1980.

Just like the EBM and vector space model, this **model** also tries to overcome the rigidity of the BM by giving document a relative relevance with respect to a query. With this **model**, information items are ranked according to the extent to which they match a given query or of how much relevance they are matched to a given query. Relevant query can be achieved by a kind of weighing process e.g index term weight, query term weight etc.

Relative relevance can also be achieved by fuzzy set theory to extend BM to a weighted system. In **fuzzy retrieval**, therefore, an item is no longer relevant or irrelevant to a certain degree. On the other hand, the use of the fuzzy model increases the complexity of the retrieval system.

In the full text system, the model provides an algorithm that calculates the weight. The document to be retrieved will also be weighed in accordance with how many times the term is mentioned in the document. The algorithm then matches rank in the query to rank in the retrieval index.

The **theory of fuzzy** set is a generalisation of classical set theory for modelling vagueness and uncertainty. The vagueness can be modelled using a **fuzzy framework**, such that with each term is associated a **fuzzy** set and each document has a degree of membership in this **fuzzy set**. This interpretation provides the foundation for many models for information retrieval based on **fuzzy theory**.

The notion of **fuzzy** sets provides a convenient tool for representing classes whose boundaries are not well defined. The key idea here is to introduce the notion of a degree of membership associated with the elements of a set. This degree of membership varies from 0 to 1 and allows modelling the notion of marginal membership.

That is to say that the **Fuzzy** set is a function that maps a value, which might be a member of a set, to a number between zero and one, indicating its actual degree of membership. A degree of zero means that the value is not in the set and a degree of one means that the value is completely representative of the set.

In-text Question

Briefly explain Vector Space Model

Answer

The Vector space model was developed to overcome the rigidity of Boolean model.Proposed in 1970, it is a statistical retrieval model that represents the documents and queries as vectors

in a multidimensional space, whose dimensions are the terms used to build an index to represent the documents. The creation of an index involves lexical scanning to identify the significant terms, where morphological analysis reduces different word forms to common stems, and the occurrence of those stems is computed.

3.0 Study Session Summary and Conclusion

Another two types of **Information Retrieval Model** have been explained in this study session. They are **Vector Space Model** and **Fuzzy Model**. Tutor marked assignment was given to test the students on their understanding of both models. Self-Assessment question were provided for students' practice.

4.0 Self-Assessment Questions

1. Write short note on Vector Space Model of information retrieval. Give advantages and disadvantages.
2. Write short note on Fuzzy Model of information retrieval. Give advantages and disadvantages.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2TqRNsu. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2HwVeai and critique it in the discussion forum

6.0 References/Further Readings

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vanRisjbergen, C.J. (2004). *The Geometry of Information Retrieval*, Cambridge UP.

Study Session 4

Probabilistic Model and Natural Language Model

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Probabilistic Model
    2.2- Natural Language Model
        2.2.1- Natural Language Search
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 4, in this session, will discuss the last two IR models in this module. They are Probabilistic Model and Natural Language Model Boolean.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Explain Probabilistic Model
2. Explain Natural Language Model

2.0 Main Content

2.1 Probabilistic Model

The *Probabilistic retrieval* model is a

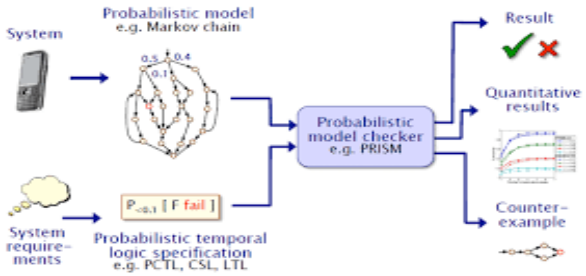

Fig 2.3.1: Probabilistic Model

statistical model proposed in 1976 based on the Probability Ranking Principle, which states ***that an information retrieval system is supposed to rank the documents based on their probability of relevance to the query, given all the evidence available***.

The principle takes into account that there is uncertainty in the representation of the information need and the documents. There can be a variety of sources of evidence that are used by the probabilistic retrieval model, and the most common one is the statistical distribution of the terms in both relevant and non-relevant documents.

In probabilistic model, you start with ***information needs***, which they translate into ***query representations***. Similarly, there are *documents*, which are converted into ***document representations***. Based on these two representations, a system tries to determine how well documents satisfy information needs. In the Boolean or vector space models of IR, matching is done in a formally defined but semantically imprecise calculus of index terms. Given only a query, an IR system has an uncertain understanding of the information need.

Given the query and document representations, a system has an uncertain guess of whether a document has content relevant to the **information need**. However, probability theory provides a principled foundation for such reasoning under uncertainty in order to estimate how likely it is that a document is relevant to an information need.

The main disadvantage of the probabilistic model according to Hiemstra (2000) is that it only defines a partial ranking of the documents. For short queries, the number of different subsets will be relatively low. By looking at a term as a definition of a set of documents, the probabilistic model ignores the distribution of terms within documents. In fact, one might argue that the probabilistic model suffers partially from the same defect as the Boolean model. It does not allow you to control the retrieved set of documents. However, probabilistic retrieval model

has the same general characteristics (**advantages and shortcomings**) as the other statistical retrieval models

In-text Question
What do you understand by Probabilistic retrieval model?

Answer
Probabilistic retrieval model is a statistical model proposed in 1976 based on the Probability Ranking Principle, which states that an information retrieval system is supposed to rank the documents based on their probability of relevance to the query, given all the evidence available.

2.2 Natural Language Model

This is a model that tries to overcome the rigidity of BM by allowing you to express his query by its everyday natural language (**spoken language**). The search engine takes each word and matches it against the word in the document store.
In principle, information storage and retrieval is simple.

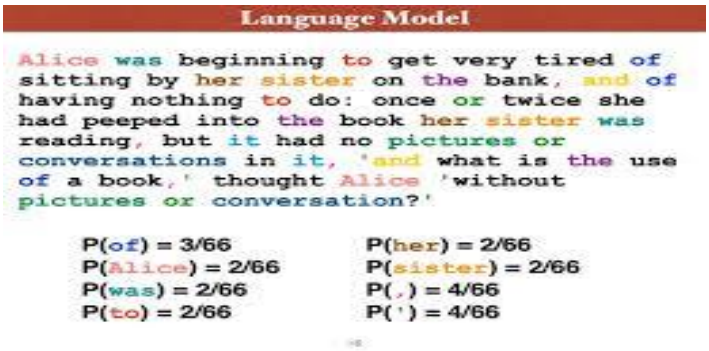

Fig 2.3.2: Natural Language Model

Suppose there is a store of document and a user formulate a question to which the answer is a set of document satisfying the information need expressed by the question. You can obtain the text by reading all the document in the store retaining the relevant ones and discarding the rest. In a sense, this constitute "perfect retrieval", however, this situation is obviously in practice. A user (you) either does not have the time or does not wish to spend the time reading the entire document collection. Apart from this fact, it may be physically impossible for him to do so.

When high speed computers became available for non-numerical work, we found out that many taught that computer will be able to "read" an entire document

collection to extract relevant document. It soon became obvious that using natural language text document not only cause input and storage problem but also led unsolved the intellectual problem of characterising the document content.

In natural language retrieval, a computer software attempt to automatically characterise document by duplicating the human process of "reading". This is a problem because reading involves extracting information both syntactic and semantic from the text using it to decide whether each document is relevant or not to a particular request. The difficulty therefore, is not only knowing how to extract the information but also how to use it to decide relevance. To date this problem has remained largely unsolved. Intellectually, it remains possible for a human to establish the relevance of a document to a query.

For a computer to do this requires the construction of a model within which relevance decision can be quantified and the pure natural language model has to a large extent fail in this regard.

Major problems of Natural Language Model

1. Problem of homonyms -two words that are spelled the same and sound the same but have different meaning. e.g Address could men 'to speak to' or a location. Also Arm could mean a body part or division of a company.
2. Problem of synonyms - word or phrase that means exactly or nearly the same as another word or phrase in the same language, for example shut is a synonym of close.
3. The full text of the document must be available so that the user's query can be matched.
4. Retrieval takes a long time because the search engine has to search for every word and compare them with the words used in the query. It is also very expensive to implement because the full text has to be in digital form
5. There is a lot of stop words which they ignore when searching. Stop words are words with little meaning that are removed from the index and the query (**they can be function words, low content words, very high frequency**

words). Almost all IR applications remove these stop words. This usually increases system performance. But there are many that are handled poorly after the removal of stop words.eg "***to be or not to be***" this phrase may be as well very important to the user which the IRS might have removed

2.2.1 Natural Language Search

Usually, '**natural language search**' means that the user (you) only has to type a request and the system takes care of **automatic query formulation**. Instead of the Boolean operators, these systems often use related operators that are easier to understand by non-expert users. The use of these operators is however not mandatory, making it possible to enter a request as shown in the example of table 5 below.

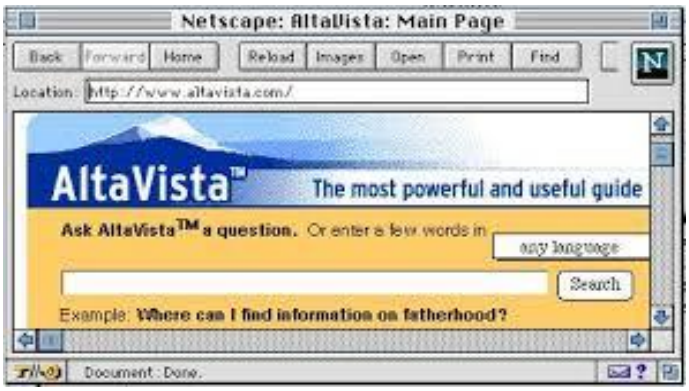

Fig 2.3.3: Natural Language Search

Table 2.4.1: Natural language search operators

Query	Interpretation
how to promote sustainable consumption patterns	real natural language request: rank the documents containing one or more of the terms
how to reduce the production of+"harmful materials" –uranium	rank the documents considering that documents should contain the phrase "harmful materials" and should notcontain the term "uranium'
why (forbid prohibit ban) wasteful packaging[0.9] of products[0.1]	rank the documents considering that the terms "forbid", "prohibit" and "ban" are synonyms and considering that "packaging" is much more important than "products"

Copied from Hiemstra (2000)

The operators which can be found in some Information Retrieval Systems such as Dialog Target, Lexis-Nexis Free Style or Altavista are summed up below Hiemstra (2000):

Exact match operator / mandatory terms

The mandatory term operator can be used to indicate that a term ***must*** be present in the selected documents. It is inspired by the AND operator in Boolean queries, but has slightly different semantics. Unlike the AND operator, which is a binary operator requiring two arguments, the mandatory term operator is a unary operator requiring one argument. The example uses the plus symbol to flag mandatory terms, but other conventions are also used, e.g. using the asterisk-character, or using a separate user interface field.

Exclusion operator

The **exclusion operator** can be used to indicate that a term should *not* be present in the selected documents. Obviously it is inspired by the NOT operator in Boolean queries. This **operator** is not as common as the exact match operator, because the absence of a term is not as clear an indication of relevance as the presence of a term. In table 5, the minus symbol in query column row 2 is used to flag terms that documents should not contain.

Synonyms and wildcards

Operators for synonyms and wildcards are inspired by the Boolean OR. Usually the operator for wildcards used for natural language queries does not differ from the operator used in **Boolean queries**. Explicit marking of synonyms is sometimes supported by putting **synonyms** between parentheses. The system uses this information to produce a better ranking. Other conventions leave the main term outside the parenthesis as in child (**minor, infant**).

Phrases

Explicit marking of **phrases** is inspired by the Boolean ADJ operator. The system uses the phrase to produce a better ranking. Identifying phrases is very useful in combination with the exact match operator to perform a high-precision search,

looking for an exact phrase or an exact quotation. Most **query languages** use single or double quotation marks to mark phrases.

Manual term weighting

Query term weights are, one way or the other, used in many ranking **algorithms**. Some systems give the user access to these weights so they can indicate themselves which terms are important and which terms are not important. The last row in table 5 gives an example of this use of term weights. Manual term weighting is for instance supported by Microsoft Index Server.

In-text Question

What are the Major problems of Natural Language Model?

Answer

1. Problem of homonyms -two words that are spelled the same and sound the same but have different meanings e.g Address could men 'to speak to' or a location. Also Arm could mean a body part or division of a company.

2. Problem of synonyms - word or phrase that means exactly or nearly the same as another word or phrase in the same language, for example shut is a synonym of close.

3. The full text of the document must be available so that the user's query can be matched.

4. Retrieval takes a long time because the search engine has to search for every word and compare them with the words used in the query. It is also very expensive to implement because the full text has to be in digital form

5. There is a lot of stopwords which they ignore when searching. Stopwords are words with little meaning that are removed from the index and the query (they can be function words, low content words, very high frequency words).

3.0 Study Session Summary and Conclusion

Probability and Natural **Language Models** were discussed in this study session. Tutor marked assignment was given to test the students on their understanding of both **models**. Self-Assessment question were provided for students' practice.

4.0 Self-Assessment Questions

1. Explain Probabilistic Model. State the Probability Ranking Principle.

2. Explain in details Natural language models. Give advantages and disadvantages.
3. With relevant examples, explain Natural Language search.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2SGOJ79. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2jHWJFQ and critique it in the discussion forum

6.0 References/Further Readings

Croft, W.B. & Lafferty, J. (2003).*Language Modeling for Information* R*etrieval*.Springer.

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Module3

Query and Query Negotiation

Contents:

Study Session 1: Query – an overview
Study Session 2: Query structure
Study Session 3: Query Negotiation

Study Session 1

Query –An Overview

Section and Subsection Headings:

Contents

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    2.1- What is a Query?
    2.2- Types of Query
        2.2.1- Navigational search query
        2.2.2- Informational search query
        2.2.3- Transactional search query
        2.2.4- Action query
    2.3- Methods of Querying
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to module 3 of this course, our first session, we shall consider what constitutes a **query**. **Query** enables individuals to interact with Systems to elicit information to address problems/issues at hand. This session will introduce you to query generally.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Explain what constitutes a query
2. Establish appropriate type of query that will facilitate easy retrieval of information
3. Discuss briefly different methods of querying an IR system.

2.0 Main Content

2.1 What is Query?

In general, a **query** is a question, often required to be expressed in a formal way. The word derives from the Latin quaere (**the imperative form of quaerere, meaning to ask or seek**). A query is a question often expressed in a formal way. It enables you to interact with systems to elicit information to address problems/issues at hands. Queries can accomplish a few different tasks.

Fig 3.1.1: What is Query?

Primarily, you can use to source for specific data by filtering (**sorting**) specific criteria. Queries can also be used for summation as well as automate data management tasks. It is a very useful tool when it comes to databases and often called by users (you) through a form. In the field of information retrieval, a query

is what you use as a search engine or database enters. A query can be a broad or a narrow query.

Broad query is defined as a **query** in which the terms are more general than the terms in the search request or task. Example, for the search task: How much blood, on average, goes through the heart in one minute? The query "heart structure" is a broad query. Again, if only half or less than half of the aspects of a multi-aspect task are included in the **query**, the query is defined as broad. In practice, the broad queries usually require the user to browse through the results or refine the query in order to find an answer to the question or to improve the relevance of the retrieved documents. This is called query expansion.

On the other hand, you can narrow a query in which the terms are specific or precise to the terms in the search task. Narrow or precise queries are targeted at finding relevant documents immediately without needing to navigate. Here, the aim is to maximise the relevance of the retrieved documents. For a query to be defined precise in a multi-aspect task, all of the main aspects of the task need to be covered in the **query**.

Let us take this example, for the task: How large were the economic losses in the September11th 2001 terrorist attack directed at Pentagon in the USA? The aspects are "**economics", "losses", "2001 terrorist attack", "Pentagon in the USA**". If only terrorist attack to Pentagon is present in the query, the query is considered imprecise or broad. Sometimes the aspects are presented in the query incompletely.

Every query is represented in the Query panel as a tree consisting of several nodes. A valid query includes at least 2 nodes:

1. the root node of the tree, and
2. a concept node

The root node does not play any particular role in the query and is not editable. The concept node, however, constitutes the foundation upon which the query will expand and is essentially the root of the query (**but not the root of the tree**).This concept node represents a specific concept that we wish to search for. More specifically, we are interested in retrieving objects present in the knowledge base whose type is determined by the concept described in the concept node.

In-text Question
What is a Query?

Answer
A query is a question often expressed in a formal way.

2.2 Types of Query

Queries are very useful tools when it comes to databases and they are often called by the user through a form. They can be used to search for and grab data from one or more of your tables, perform certain actions on the database and even carryout a variety of calculations depending on your needs. The following are the common types of **query**:

2.2.1 Navigational search query is a search query entered with the intent of finding a particular website or webpage. Using navigational search query implies that you have a premeditated site in mind. This query reduces the total number of results.

2.2.2 Informational search query are queries that cover a broad topic for which the response rate is very high. When informational search query is entered into **Google** or another search engine, it implies information is sought for – hence the name of the **query**. This is used when there is the need for information.

2.2.3Transactional search query is a **query** that indicates intent to complete a transaction such as making a purchase. This may involve exact brands and product names.

2.2.4Action query is a query that makes changes to or moves several records in a simple operation. There are four (4) types of action queries; append, update, delete and make-table query.

Append query: take the set result of a query and "**appends**" (adds) to them to an existing table.

Delete query: implies that all records on a table should be deleted.
Make table query: as the name suggests implies that a table be created to reflect the set results of a **query**.

Update query: This is an action query that changes set of records according to criteria (**search conditions**) specified. It is used for modifying existing records in a table.

2.3 Method of querying

There different methods a user can adopt in querying an IR system depending on its conveniency. The following are the common methods of querying:

Browsing is going through the content of a document store in an unstructured, haphazard manner without applying any formal type of query.

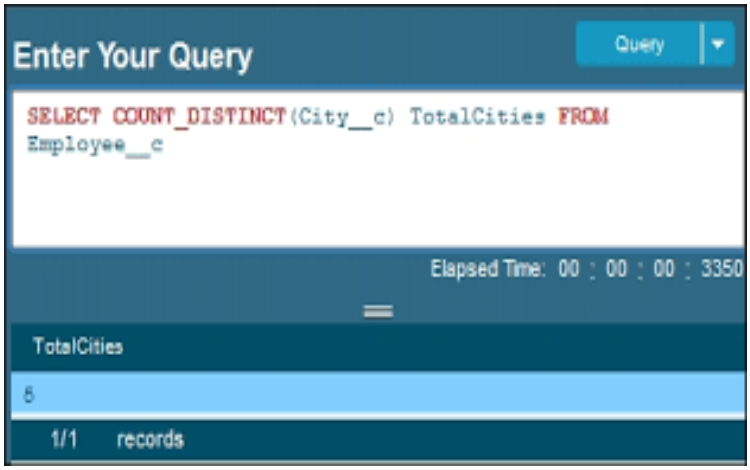

Fig 2.1.2: Method of querying

Adhoc Retrieval: A user (you) tries to find the documents that are relevant to his search using unplanned ADHOC queries. In other words, the queries are unstructured.

Structured Query: The user (you) identifies his/her information need, breaks his need into search terms. Combine these terms together using relevant operators where necessary before going to the document store or using the IR system. The user plans his search offline before executing it online.

In general queries are used to find items relevant to an information need. Queries can be regarded virtual items (not real or tangible- you cannot lay your hands on it), they are things that exists in the mind of the users (you). The aim of an IR system is to find real item (real or tangible – which user can lay hands on) that are as similar as possible to the virtual information item.

In-text Question
Mention the three methods of querying

Answer
1. *Browsing*
2. *Adhoc Retrieval*
3. *Structured Query*

3.0 Study Session Summary and Conclusion

This study session has clearly introduced student to the meaning of query, querying, types and method of querying. It has established the fact that query is a virtual item which when posed to the IR system is capable of getting the real item relevant to the information needs of the user.

4.0 Self-Assessment Questions

1. Explain what you understand by a query
2. Enumerate and discuss the different types of query
3. Explain the different methods used in querying

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://bit.ly/2tPPAs1. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/30Ap7hv and critique it in the discussion forum

6.0 References/Further Readings

vanRisjbergen C.J. (2004). The Geometry of Information Retrieval. Cambridge UP.

Voorhees, E.M. & Harman, D.K. (2005). TREC: Experiment and Evaluation in Information Retrieval. MIT Press.

Study Session 2

Query Structure

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Structure of Queries
        2.1.1- Strong Query Structures
        2.1.2- Weak Query Structures
    2.2-Query Formulation and Expansion
        2.2.1- Query Formulation
        2.2.2- Query Expansion
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 2, in this session, we shall consider the discrete structures of queries. **Query** formulation and expansion will be extensively addressed here.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. explain the structures of queries
2. differentiate between strong and weak query structures
3. identify the factors affecting query formulation

2.0 Main Content

2.1 Structure of Queries

Query structure means the use of operators to express the relations between search keys. The selection of good search keys is difficult but crucial for good search results. It is important for you to note that, the structure of queries may be described as weak (**queries with a single operator, no differentiated relations between search keys**) or strong (queries with several operators, different relationships between search keys). You should equally note that strong structured queries are known to perform better. More precisely, strong query structures are based on facets. Each facet indicates one aspect of the information request or task, and is represented by a set of concepts, which, in turn, are expressed by a set of search keys.

Fig 2.2.1: Structure of Queries

The main attribute of **query structures** lie in the weightings of the search keys. Several operators are used to indicate several query structures. For instance, weak query structures are indicated by the following:

1. SUM, which is an unexpanded query, means average of the weights of search keys
2. WSUM1, which is an expanded query of SUM, means weighted average of the weights of keys, and

3. WSUM2, another expanded query of SUM, means weighted average of the weights of keys.

On the other hand, strong query structures are indicated by the following:

1. BOOL , a query with Boolean operator, which is a facet based query structure
2. SSYN, a SUM-of-synonym-groups-query; here, each facet formed a clause with the SYN operator
3. ASYN, similar query to SSYN queries, but with SYN groups combined with the probabilistic AND operator, ie. a product of facet was calculated instead of an average
4. XSUM, a modification of the Boolean query structure: facets were combined with SUM operator instead of AND operator
5. OSUM, is a combination of Boolean queries consisting of term keys of different facets and SYN clauses consisting of expansion keys for each facet.

2.1.1 Weak Query Structures

The weak **query structures** used to combine the search keys of the above search request can be represented by the SUM, WSUM1 and WSUM2 queries. An unexpanded SUM query can be constructed by the system from the original concept of the request, and each concept is represented by a single key or a set of keys corresponding to the term but without phrases.

The unexpanded query structure is as follows:

SUM

#sum(nuclear power plant nuclear station atomic power plant atomic reactor nuclear reactor radioactive waste nuclear waste storage store risk accident danger hazard disaster catastrophe)

In the **query expansion**, all expressions will be as single words, ie. no phrases were included. The original keys were usually weighted higher than the expansion keys. Thus in WSUM queries expansion keys and the keys of equivalent

expressions were given smaller weights than the keys of the original concepts. A typical WSUM1 query structure will be given as follows:

WSUM1

#wsum(1 2 #3(nuclear power plant) 1 #3(nuclear station) 1 #3(atomic power plant) 1 #3(atomic reactor) 1 #3(nuclear reactor) 2 #3(radioactive waste) 1 #3(nuclear waste) 2 storage 1 store 2 risk 2 accident 1 danger 1 hazard 1 disaster 1 catastrophe)

In WSUM2, query keys will be weighted according to the type of expansion they belonged to. A typical WSUM2 query structure will be given as follows:

WSUM2

#wsum(1 10 #3(nuclear power plant) 9 #3(nuclear station) 9 #3(atomic power plant) 7 #3(atomic reactor) 7 #3(nuclear reactor) 10 #3(radioactive waste) 9 #3(nuclear waste) 10 storage 9 store 10 risk 10 accident 9 danger 9 hazard 7 disaster 7 catastrophe)

2.1.2 Strong Query Structures

Similarly, queries can be constructed from the search request of above using the various operators representing the strong query structures. Using the same query request, the strong query structures will be the following:

SSYN

#sum(#syn (#3(nuclear power plant) #3(atomic power plant)
#3(nuclear station) #3(atomic reactor) #3(nuclear reactor))
#syn(#3(radioactive waste) #3(nuclear waste))
#syn(storage store)
#syn (risk accident danger hazard disaster catastrophe))

ASYN

#and (#syn(#3(nuclear power plant) #3(atomic power plant) #3(nuclear station) #3(atomic reactor) #3(nuclear reactor)
#syn (#3(radioactive waste) #3(nuclear waste))
#syn (storage store)
#syn (risk accident danger hazard disaster catastrophe))

XSUM

#sum (#or(#3(nuclear power plant) #sum(#3(atomic power plant)
#3 (nuclear station) #3(atomic reactor) #3(nuclear reactor)))
#or (#3(radioactive waste) #sum(#3(nuclear waste)))
#or (storage #sum(store))
#or (risk accident #sum(danger hazard disaster catastrophe)))
OSUM
#sum (#and(#3(nuclear power plant) #3(radioactive waste) storage
#or(risk accident))
#syn (#3(atomic power plant) #3(nuclear station) #3(atomic reactor)
#3 (nuclear reactor))
#syn (#3(nuclear waste))
#syn (store)
#syn (danger hazard disaster catastrophe)

In-text Question
What is the mean of structure of a Query?

Answer
Query structure means the use of operators to express the relations between search keys.

2.2 Query Formulation and Expansion

In **query formulation**, generally three levels can be differentiated; - a conceptual, linguistic and string level.

Fig 2.1.2: Query Formulation and Expansion

To demonstrate the different levels of query **formulation and expansion**, let us assume that the following is a query task or request: Storage of radioactive waste produced in nuclear power plants, examples of risks and accidents. At the conceptual level, about four facets and seven concepts can be

recognised from this request. Therefore, a typical query "plan" can be as follows: nuclear power plants AND radioactive waste AND storage AND (risk OR accident)

At the string level, terms are replaced by search keys. The query is expressed with syntax of the query language (**the linguistic level**). So sample query is first formulated into the Boolean query structure (**depending on the information retrieval model used**). An unexpanded query will contain the search concepts selected on the basis of the request and given thus:

#and(#3nuclear power plant) #3(radioactive waste) storage #or(risk accident)
With query expansion, synonyms of the terms are added to the query. The sample query with the synonym expansion is the following:
#and(#or(#3nuclear power plant) #3(nuclear station) #3(atomic power plant))
#or(#3(radioactive waste) #3(nuclear waste))
#or(storage store)
#or(risk accident danger hazard))

2.2.1 Query Formulation

Your **query formulation** is simply the process by which a user or searcher defines his/her information needs. This means that in every **query formulation** technique there is a human in the loop. From very simple or narrow queries to extremely complex or broad queries, there must be a person to define the information need in the form of a query. **Query formulation** is an essential part of successful information search and retrieval. It is typically based on search keys given by the user and is a major step in the complex process of information search. It therefore poses a huge challenge to users to formulate effective queries for their web information search. This is more so, given that the web is used by a diverse population varying in their levels of expertise. Your information search consists of four main steps: problem identification, need articulation, **query formulation**, and results evaluation. This process of information search is affected by numerous factor including, environmental (e.g., the database and the search

topic), user or searcher (e.g., **online search experience**), search process (e.g., **commands used**), and search outcome variables (e.g., **precision and recall**).

Factors Affecting Query Formulation

It is known from studies that three main factors affect query formulation:

1. Media expertise, including familiarity with the search environment, search engine expertise, computer expertise and expertise in information retrieval
2. Domain expertise
3. Type of search task.

In media (web) expertise, the more experienced web user the searcher is, the more likely he/she is to use a "**straight to information**" search style (or **narrow query**) rather than a broader "**navigating to information**" style (or broad query). We observe that the more experienced the user becomes with the web, the more the search engines become a tool, rather than something to spend time with. Thus, the users (you) want to search information as efficiently as possible and not just to see if something interesting happens to come by. The web experience also makes the users understand the structure of the material and the usual style of writing in the web. These are important skills for successful information search.

Domain expertise presumably helps people in **query formulation** by giving them a possibility to use either more terms in their queries (synonyms), or possibly fewer, but more accurate terms. Thus, domain expertise is not directly expected to lead to longer queries, but the quality of the selected terms is expected to be high.

Search task is divided into three broad categories: fact-finding, exploratory, and comprehensive search tasks. In fact-finding, the source of information is not a key issue, but precision of the result set is a key issue for efficient search. On the other hand, in exploratory search tasks, the searcher's aim is to obtain a general idea of the search topic or possibly to retrieve a couple of documents as an example. So here, high precision of the result set is not necessarily the most important thing. When the task is to find as many documents as possible on a given topic, then

comprehensive search task is indicated. In this case, the recall should be as high as possible for the search to be successful.

2.2.2 Query Expansion

It is important for you to note that, **query expansion** is the process of reformulating a seed query to improve retrieval performance in information retrieval operations. In the context of web search engines, query expansion involves evaluating a user's input (**what words or other types of data were typed into the search query area**) and **expanding** the search **query** to match additional documents. This is because the original query does not always give satisfactory results. Typical **query expansion** involves techniques such as:

1. Finding synonyms of words, and searching for the synonyms as well
2. Finding all the various morphological forms of words by stemming each word in the search query
3. Fixing spelling errors and automatically searching for the corrected form or suggesting it in the results
4. Re-weighting the terms in the original query.

It has shown us that many information users or searchers are not always clear about what they are looking for and even if they are, they are not always sure about how to **formulate that information need**. In information retrieval systems, one of the ways in which this can be accommodated to some degree is allowing the user to expand the initial query (**query expansion**), possibly after some relevant documents have been found, in order to enrich the query specification.

Your purpose of **query expansion** is to make the query resemble more closely the relevant documents and thus, to retrieve those relevant documents. Therefore, **query expansion** could mean adding or deleting terms from the original query or even changing terms. You can do this by using information from relevance feedback with relevant documents identified manually by the user or by assuming the top-ranked documents from an initial ranking are relevant.

In-text Question
What are the three factors affect query formulation ***Answer*** *1. Media expertise, including familiarity with the search environment, search engine expertise, computer expertise and expertise in information retrieval* *2. Domain expertise* *3. Type of search task.*

3.0 Study Session Summary and Conclusion

This study session has exposed students to the meaning of **query structure** including what constitutes the weak and strong query structures. How queries are **formulated and expanded** were also discussed. The study session was concluded with the factors affecting **query formulation**. Tutor marked assignment and self-assessment questions were given to students for practice

4.0 Self-Assessment Questions

1. Explain weak and strong queries with relevant examples
2. Enumerate and discuss the different factors affecting query formulation
3. Write short note on query formulation and query expansion.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://bit.ly/2EJntRR. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/30EtqbG and critique it in the discussion forum

6.0 References/Further Readings

vanRisjbergen C.J. (2004). The Geometry of Information Retrieval. Cambridge UP.

Voorhees, E.M. & Harman, D.K. (2005). TREC: Experiment and Evaluation in Information Retrieval. MIT Press.

Study Session 3

Query Negotiation

Section and Subsection Headings:
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    2.1- What is Query Negotiation?
    2.2- Reference Interview
        2.2.1- Parts of Reference Interview
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 3, in this session, we shall be made to understand what **query negotiation** entails. Reference Interview which is a form of **query negotiation** shall be discussed extensively here.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. explain what we mean by query negotiation
2. explain reference interview and its components

2.0 Main Content

2.1 What is Query Negotiation?

Natural language designates a method and means that people use to express ideas and communicate. Because subjective is a significant part of a communication process, information is inferred when we communicate. Thus, the meaning of an expression will be changed completely when it is taken out of the context. A major problem in information retrieval results from users subjectively and fuzzy expressions.

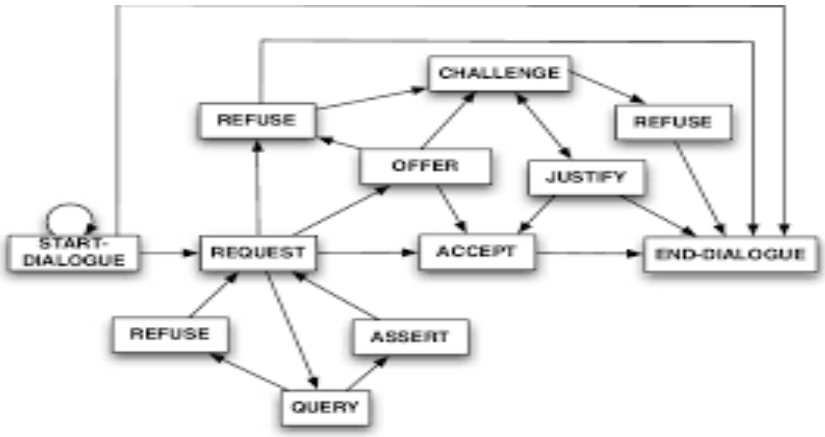

Fig 2.3.1: What is Query Negotiation?

Any query results in either finding or not finding the correct information to meet a user's demand. The process of **query negotiation** is that of translating the user's need into a series of search expression. Typically a document query is more general than a data query. A user's (your) perception of what his/her needs are changes in the course of his/her search that is why in most cases an information expert is being call upon to assist a user. An example of query negotiation is reference interview which will be discussed in the next section.

Query negotiation is the interaction between a person with an information need and information systems. Its purpose is to refine information need so that it can be usefully responded to by the information service (system).**Query Negotiation** aims at identifying and satisfying users' information needs. This includes a series of decision making steps: from the receipt of a user's query to communication of an answer and/or direction to appropriate tools that may satisfy the needs. This requires effective communication skills. **Negotiate** the subject matter component of the query first.

Query negotiation guidelines

1. Use open-ended questions at the initial stage of the negotiation. For instance, it can state with; when, where, how, who, what
2. Used closed question towards the end of the negotiation, as appropriate
3. This would lead to either Yes or No answers

In-text Question
Define Query negotiation

Answer
Query negotiation is the interaction between a person with an information need and information systems.

2.2 Reference Interview

A type of **query negotiation** is Reference interview (RI). Reference interview is the ability by skilful questioning without appearance of curiosity or impertinent to elicit information from the vaguest most general request to a clear idea of what the enquirer or user really needs. It is a means or facility of reducing large abstract demands to concrete search terms.

Fig 2.3.2: Reference Interview

Human professional acts as intermediary between the user and the **query**. This is to help the user to be able to express his/her information need. Conducting a reference interview is the most important work a reference librarian does. This is because this enables matching of patron's question to relevant and useful information sources. The components of reference interview are listed below

Components of Reference Interview

1. The user's information need
2. The format of the information needed
3. The setting in which the interview will take place i.e on-line or manual
4. History of the user's attempt to search for the required information
5. How much information is desired?
6. Level of treatment of information from trivial to scholarly etc.

As an information professional, you need a very good communication skill to carry out a reference interview. A reference interview can be conducted by an electronic machine. This is known as e-reference interview. It does not require the enquirer to be physically present while the **enquiry** is taking place. This type of interview has some specific disadvantages as enumerated below:

1. The loss of non-verbal cues- tone of voice, facial expression e.t.c cannot be detected by electronic machines. So user's express frustration with this unlike when the user is interacting physically with a human.
2. There is a phenomenon of "disappearing questioner". Computers never respond to questions unlike human. When there is interaction between human, questions can be asked again and again for clarification.
3. A longer time is needed (i.e to clarify, to break down)

In summary, during the process of **query negotiation**, the information user perceives most of their information need. Sometimes, information professional can help define what is really needed as some information users may not know how best to express their information need. An information professional is able to detect the actual need sometimes through body or facial expression of the information user. Information professional can also provide how best to meet the need of the user using available technology.

The information user (you) has to set priorities based on ultimate objective and the time, labour and money available for searching. Together, the information professional and the user evolve the strategy needed for extracting required information from a document store.

2.2.1 Parts of Reference Interview

The parts of Reference Interview are enumerated and discussed below:

1. Open the interview
2. Negotiate the question (as needed)
3. Search for information
4. Communicate the information to the user
5. Close the interview

Open the interview commences with acknowledging users' presence. Express interest the user query. It is very important you listen carefully to what the user is saying. Reflect on the message of the query. For instance, what is the subject matter of the query? What kind of query is it (**factorial, research** etc). Similarly, it is important to decide whether to negotiate or not **negotiate** the query. Sometimes, no **negotiation** is required but only probing a bit may clarify the need. If the **query message** is clear and little or no **negotiation** is needed then it becomes imperative to identify appropriate types of answer-providing tools; identify titles without the tools to use, obtain user feedback. If information need is met, then the interview is concluded.

Negotiate User Question: When your **query** is vague, incomplete, inaccurate, missing pieces of **information** or outside the mediator's knowledge domain, then negotiation becomes necessary. Thus probing is mandatory to identify the user's true information needs.

Search for information: match user query to appropriate types of Advanced Threat Protections (ATPs)

1. identify specific sources within appropriate ATPs
2. describe process of information search to the user
3. If the user is physically present, instruct in the use of sources as appropriate
4. evaluate the search process

Communicate information to the User: Communicate the relevant information to the user (you). This can be performed relevant web pages to the library user if the

library provides digital reference services. Also, email URLs with relevant information. Lastly, avoid overwhelming users with information.

Close the interview: Mediator may close the interview when patrons have enough information to work with or when request is met and user is satisfied. A user may close the interview by thanking the mediator. Users should be encouraged to come for assistance whenever the need arises.

In-text Question
Itemize the components of Reference Interview

Answer
1. *The user's information need*
2. *The format of the information needed*
3. *The setting in which the interview will take place i.e on-line or manual*
4. *History of the user's attempt to search for the required information*
5. *How much information is desired?*
6. *Level of treatment of information from trivial to scholarly e.t.c*

3.0 Study Session Summary and Conclusion

This study session has exposed students **query negotiation** and reference interview as a very good example of **query negotiation**. Tutor marked assignment was given to students. Self-assessment questions were also given to students for practice.

4.0 Self-Assessment Questions

1. What is query negotiation?
2. Give detailed explanation of what you understand by reference interview
3. What are the disadvantages of a computer carrying out a reference interview?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2Vtn6QO. Watch the video & summarise in 1

paragraph

b. View the animation on add/site https://bit.ly/2HKpLAk and critique it in the discussion forum

6.0 References/Further Readings

vanRisjbergen C.J. (2004). The Geometry of Information Retrieval. Cambridge UP.

Voorhees, E.M. & Harman, D.K. (2005). TREC: Experiment and Evaluation in Information Retrieval. MIT Press.

Module 4

Information Retrieval Process and Evaluation

Study Session 1: Information Retrieval Process- an overview
Study Session 2: Search Strategy
Study Session 3: Evaluation Retrieval Products
Study Session 4: Standard Metadata and their description

Study Session 1

Information Retrieval Process- An Overview

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Information Retrieval Processes – an overview
    2.2- Optimizing the Retrieval Process
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to module 4, from the above previous discussion, I can deduce that you have gained so much of what this course offered you.

Our primary concern in this session is the basic processes of **information retrieval**. With the aid of a diagram, the activities were clearly explained. Four strategies of optimising a **retrieval process** were also discussed in this study session.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Explain Basic Information Retrieval Processes
2. Optimise a Retrieval Process

2.0 Main Content

2.1 Information Retrieval Processes- An Overview

The main activity of an **Information Retrieval System** (IRS) is the gathering of the relevant documents that best satisfy user information needs or requirements (**queries**). Both documents and user queries must be properly represented in a reliable way, so that IRS can satisfactorily develop the retrieval activity.

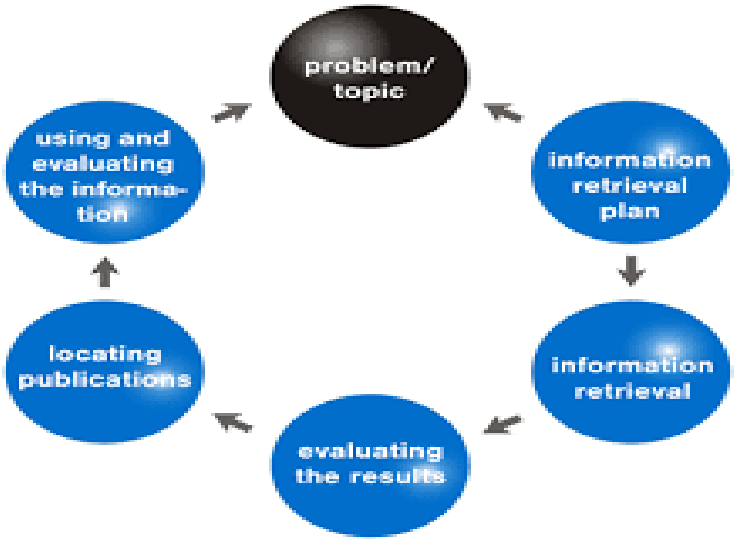

Fig 4.1.1: Information Retrieval Processes- an overview

Basically, IRSs present three components to carry out their activity:

1. A Data base:which stores the documents and the representation of their information contents (**index terms**). It is built using tools for extracting index terms and for representing the documents.

2. A Query Sub system:This allows users to formulate their queries by means of a query language.
3. An Evaluation Subsystem: This evaluates the documents for a user query. It presents an inference procedure that establishes a relationship between the user request and the documents in the database to determine the relevance of each document to the user query.

You should note that **information retrieval** is often a continuous process during which you will consider, reconsider and refine your research problem, use various different information resources, information retrieval techniques and library services and evaluate the information you find. In practice the process of information retrieval involve a user having an information need which he forms a query with. The user (you) then sends the query to a system and the system in turn returns an answer set. The user (you) then eyes and evaluates the results and if the user is satisfied, he stops but if the user is not satisfied, he modifies the query and sends the query back to the system until he finally gets the desired result.

Basic processes of information retrieval

You have three basic processes an **information retrieval system** has to support: the representation of the content of the documents, the representation of the user's information need, and the comparison of the two representations (Hiemstra, 2000). The processes are visualised the figure below (Croft 1993).

In the figure, rectangle represent data and rounded rectangle represent processes.

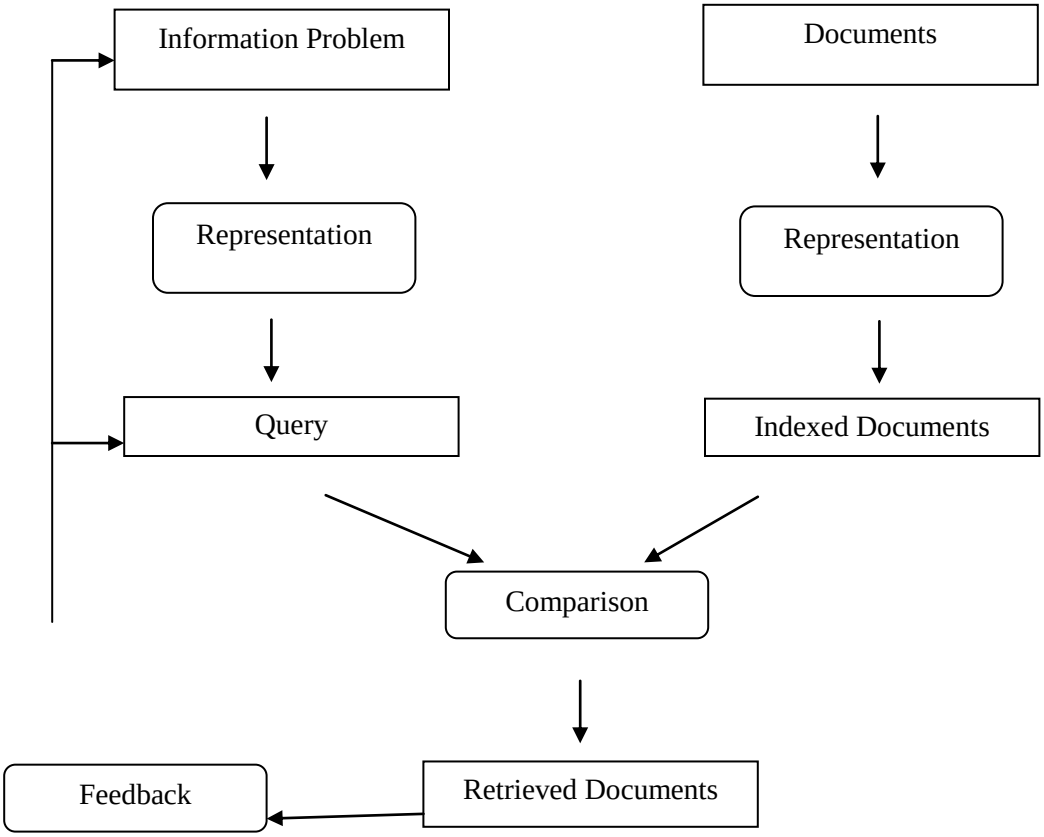

Figure 4.1.2: Information Retrieval Processes (Croft 1993)

Documents and Indexed Documents: Representing the documents is usually called the ***indexing*** process. The process takes place off-line, that is, the end user of the information retrieval system is not directly involved. The indexing process results in a formal representation of the document: the index representation or document representation. Often, full text retrieval systems use a rather trivial algorithm to derive the index representations, for instance an algorithm that identifies words in an English text and puts them to lower case.

The indexing process may include the actual storage of the document in the system, but often documents are only stored partly, for instance only title and abstract, plus information about the actual location of the document.

Information Problem and Query: The process of representing the information problem or need is often referred to as the ***query formulation* process**. The resulting formal representation is the **query**. In a broad sense, query formulation might denote the complete inter active dialogue between system and user, leading not only to a suitable **query** but possibly also to a better understanding by the user of his/her information need.

Feedback: The automatic formulation of successive **queries** is called *relevance* ***feedback***. The user and the system communicate the information need by respectively queries and retrieved sets of documents. This is not the most natural form of communication. Humans would use natural language to communicate the information need amongst each other. Such a natural language statement of the information need is called a request. Automatic **query formulation** inputs the request and output an initial **query**. In practice, this means that some or all of the words in the request are converted to query terms, for instance by the rather trivial algorithm that puts words to lower case. Relevance feedback inputs a **query** or a request and some previously retrieved relevant and non-relevant documents to output a successive **query**.

Comparison: The comparison of the query against the document representations is also called the ***matching* process**. The matching process results in a ranked list of relevant documents. Users will walk down this document list in search of the information they need. Ranked retrieval will hopefully put the relevant documents somewhere in the top of the ranked list, minimizing the time the user has to invest on reading the documents. Simple but effective ranking algorithms use the frequency distribution of terms over documents. Ranking algorithms based on statistical approaches easily halve the time the user has to spend on reading documents.

2.2 Optimising the Retrieval Process

There are a number of strategies used in optimising the retrieval process.

1. Boolean Model (BM) - Using the BM strategy in translating the search

process into query language in the database to be searched using whatever future process. Boolean operators are then used to show the relationship between the different search terms in the search strategy while at the same time. Keeping some ideas in reserve for possible modification of the search after seeing the result of the initial search statement

2. Building Block (BB) - With BB strategy the searcher starts with an initial search term and reviews the result, then continues combining with additional search terms one after the other until it satisfies with the outcome of the search. It is very similar to the BM because the terms are combined with the Boolean operators.
3. Successive Fraction (SF) - starts from the general to the particular. g it starts from the general and continues excluding irrelevant terms until you get a more precise result. In SF approach, the searcher works from general to the particular, beginning with the entire database and gradually narrowing it search into smaller and smaller fractions until to narrow the search further will begin to omit relevant records. It is not a time efficient strategy (it wastes time).
4. Citation Pearl Growing (A document of great value) - In Citation Pearl growing, a searcher begins with a key record already known "a citation pearl" of great price. This record is first retrieved from a database and then the search is extended in a number of ways which include:

    i. Using the key word ascribed to Citation Pearl to see if there are other records in the database using the same keywords
    ii. Finding other records by the author of the citation pearl on a similar or closely related subject
    iii. Finding from the corporate source field (if it exists) of the citation pearl if any other records on a similar subject have emanated from the same organisation under different authors
    iv. Finding where citation searching exists on a database (e.g scholar google.com) records that have cited this article, which authors have cited the citation pearl at the end of their papers.

In these ways (A-D), a body of records are gradually built up, all connected with the original key record. The citation pearl works in opposite direction of the successive fraction approach.

In-text Question
What are the three components IRSs?

Answer
1. A Database: which stores the documents and the representation of their information contents (index terms). It is built using tools for extracting index terms and for representing the documents.
2. A Query Subsystem: This allows users to formulate their queries by means of a query language.
3. An Evaluation Subsystem: This evaluates the documents for a user query. It presents an inference procedure that establishes a relationship between the user request and the documents in the database to determine the relevance of each document to the user query.

3.0 Study Session Summary and Conclusion

This study session has made the concept of **Information Retrieval processes** easy to understand using the diagrammatic representation of the basic processes of **Information Retrieval**. Different strategies in which information retrieval process can be optimized were discussed. Tutor marked assignment was given to test the students. Self-Assessment questions were provided for students' practice.

4.0 Self-Assessment Questions

1. With the aid of a diagram, explain the basic processes of Information Retrieval
2. Enumerate and explain clearly the strategies of optimizing the Information Retrieval process.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/1FtyP54. Watch the video & summarise in 1 paragraph
b. View the animation on add/site https://bit.ly/30DPN0C and critique it in the

discussion forum

6.0 References/Further Readings

Chowdhury, G.G (2003). Introduction to Modern Information Retrieval. Neal-Schuman

Croft, W.B. (1993).Knowledge-based and Statistical Approaches to Text Retrieval.*IEEE Expert* 8(2), 8-12

Study Session 2

Search Strategy

Section and Subsection Headings:

Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Search Strategy processes
    2.2- Factors affecting on-line search
    2.3- Tips on conducting searches
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 2, this will session described what a **search strategy** is and its importance. It also discussed the basic processes of a search strategy, factors affecting on-line search, and the tips a user can use in conducting a search for effective results.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain the search strategy processes
2. Explain the basic factors affecting on-line search
3. Use the tips for conducting a search to retrieve relevant information

2.0 Main Content

2.1 Search Strategy processes

Your **search strategy** is an overall plan for the search process. You can call it a systematic plan of carrying out a search in an IR system in order to dig up for **relevant information**. It is very important to note that developing a **search strategy** is extremely important. Not only does the strategy provide you with an overall structure for your search, it provides a record of your search history. As your **research** develops and your search is refined, your search history can be extremely useful.

Fig 4.2.1: Search Strategy processes

It can also improve the relevancy of results obtained as you have thought through your keywords and **synonyms** and how these relate to each other. A **search strategy** is an organised structure of key terms used to **search** a database. The **search strategy** combines the key concepts of your search question in order to retrieve accurate results. Each database works differently so you need to adapt your **search strategy** for each database. You may wish to develop a number of separate **search strategies** if your research covers several different areas. It is a good idea to test your strategies and refine them after you have reviewed the search results. How a user carries out a **search determines** the **quality of the information retrieved**.

The following are components required of a search process:

1. Define the scope of your search

2. Identify the search concepts
3. List keywords for each concept
4. Specify the logical relationships among keywords. This is where BM strategy comes in
5. Identify the relevant databases or search engine or file you want to use for searching
6. Build your search query and execute
7. Conduct an exploratory search
8. Evaluate your answers/ results- are they small or big, are they relevant to your information need. Evaluating your result determines whether you will refine your search. If they are relevant to your information need, then there is no need to refine your search
9. Refine your search- by dropping keywords, modifying key words e.t.c. this leads to building your queries again.

Example: How to Package given evidence of Tampering Drugs

1. Define the scope by breaking down the search topic into components. The three major topics here are: Package, Tampering and Drugs
2. The search concepts are: Package, Tampering and Drugs
3. The key words for the concepts are broken down as follows:

Concepts Keywords

a. Package, container, sachets, bottle, carton e.t.c
b. Tampering, any type of tampering e.g abuse, vandalizing e.t.c
c. Drugs , medical, surgical, veterinary e.t.c

4. Identifying the specify logical relationships among keywords e.g Package OR bottle NOT glass

   You can now follow the remaining steps above from 5-9

In-text Question

Mention seven components required of a search process

Answer

1. *Define the scope of your search*
2. *Identify the search concepts*
3. *List keywords for each concept*
4. *Specify the logical relationships among keywords. This is where BM strategy comes in*
5. *Identify the relevant databases or search engine or file you want to use for searching*
6. *Build your search query and execute*
7. *Conduct an exploratory search*

2.2 Factors affecting on-line search

Generally, the factors affecting the on-line system fall into four groups.

1. The system design factor: factors that are directly attributable to the design of IR system itself. There must be relationship between the users and the interface. Some system appears more difficult to query than others because:
   a. The language use is more complex
   b. The search logic is less straight forward
   c. More steps are involving in interrogating the system
   d. Less on-line aids are given to the user
   e. Errors are adequately difficult or more difficult to correct
   f. Greater accuracy is required in spelling, punctuation e.t.c for example AND in uppercase will be recognized as a Boolean operator. Some use full stop or asterisks to represent AND. Most retrieval system makes it interactive and close to natural language. You have to design the system to be effective.

These factors are likely to affect the tolerance of the user because if the search process is too complex or frustrating, the searcher is likely to give up in disgust, these factors will also affect the time it takes to conduct the search

2. Factors that affect the performance (response time): System response time will also be affected by the algorithm and program logic. Disk access

procedures are related factors (**evaluation from the user's point of view**)
Hardware factors: the VDU generally allows faster interaction. They are also more appealing to the majority of the users and facilitate various types of display e.g **hierarchical vocabulary display**. They are also very useful for scanning large amount of texts. Keyboard that have keys dedicated system command may be easier to use than conventional typewriter keyboard which require typing of a command name. It is important to note that number of terminals a system can support may affect the accessibility of the system to the user and that the number of the users interrogating the system concurrently will also affect the response time (higher hard disk memory, processing speed, RAM memory capacity) will help

3. Training factors: the more experienced a searcher has had with a system, the less time he will wait at the terminal and more effective his/her search will be. An experienced searcher is likely to search a system with fluency than the inexperienced or casual user is likely to.
4. Intellectual factors relating to the database itself: these factors include the characteristics of the indexing, the vocabulary or indexing language of the system. If the search strategies needed to interrogate the system is too complex, the user is likely to give up the search in disgust.

All the various factors mentioned above will influence the effectiveness and efficiency of an online retrieval system. However, in the long run, the intellectual factors are likely to have a more significant effect on the performance of the retrieval system

2.3 Tips on conducting searches

1. Read the directories at each search site
2. Check your spellings

If your results are not satisfactory, repeat the search using alternative terms.

3. Experiment with different search engines. No two search engines work from the same index.

4. Try search engines that allow you to search multiple search engines simultaneously (meta search engine)
5. If you have so many results that are not relevant, try the following:
    a. Add concept words
    b. Use vocabulary that is specific to your search
    c. Link appropriate terms with the Boolean AND
    d. If your search term is a phrase, enclose with a quotation mark (“ ”)
    e. Use the Boolean NOT to keep out records containing terms you don’t need
6. If you have few results, drop the least related terms

Fig 4.2.2: Tips on conducting searches

In-text Question
List the tips required for conducting searches

Answer
Tips on conducting searches
1. *Read the directories at each search site*
2. *Check your spellings*
3. *If your results are not satisfactory, repeat the search using alternative terms*
4. *Experiment with different search engines. No two search engines work from the same index.*
5. *Try search engines that allow you to search multiple search engines simultaneously (meta search engine)*
6. *If you have so many results that are not relevant*

3.0 Study Session Summary and Conclusion

This study session explained the **search strategies** and its processes. It also identified the factors that affects on-line search and gave relevant tips on conducting searches. Tutor marked assignment was given to test the students' ability to identify and explore more search strategy processes and factors that affects on-line searches. Self-Assessment question were provided for students' practice.

4.0 Self-Assessment Questions

1. Explain with relevant examples the processes of a search strategy
2. Explain the factors affecting on-line searches

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2tLZsTI. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2JA0M5U and critique it in the discussion forum

6.0 References/Further Readings

Baeza-Yates, R. and Ribeiro-Neto, B. (1999).*Modern Information Retrieval*.Addison-Wesley

Chowdhury, G.G (2003). *Introduction to Modern Information Retrieval*.Neal-Schuman

Study Session 3

Evaluating Retrieval Products

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- Evaluating retrieval products
    2.2- Concept of relevance
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 3, our primary concern here is to discuss how to **evaluate retrieval products** from both the users' point of view and the management point of view.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain recall and precision
2. Understand the concept of relevance

2.0 Main Content

2.1 Evaluating retrieval products

When we evaluate retrieval products, we do so from two main points of view:

1. Point of view from the system user
2. Point of view from the management

*The **system user**:* Users consciously or unconsciously evaluate the output from **retrieval system** in terms of how well it satisfies his information requirement. In other words, the users are not particularly interested in how the system works but are interested in the end result. The basis with which users evaluate the output in IRS can be measured by two concepts i.e the concept of recall and precision.

Fig 4.3.1: Evaluating retrieval products

Recall

A user is primarily concerned with obtaining the document that satisfies his/her **information need**. In such cases, one or unique document suffices. In other cases, a few key documents are what are required, while in certain situations the user is concerned with retrieving as much as possible of the total literature dealing with the subject of his inquiry. In all cases, the essential requirement is that the system must be able to retrieve documents relevant to the various subject requests. The ability of the IR system to uncover document relevant to the searcher ‘s request is known as the RECALL POWER OR RECALL ABILITY of the system.

Recall is measured by

$$\text{Recall} = \frac{\text{Number of relevant documents}}{\text{Total number of relevant document in the system}}$$

This is known as recall ratio. Sometimes it is multiplied by 100 so that it is expressed in percentage. Recall is obviously the most important requirement of the user of a **retrieval system** since his own purpose in approaching the system is to obtain more document useful in relation to his or her information need.

Precision

Figures that measure performance of an IRS are in themselves meaningless because it is possible to get 100% recall by going through the entire collection and examining every item. So the precision ratio was devised as a means of measuring the effort put into achieving recall. Precision ratio is defined as the number of relevant document retrieved over the total number of document retrieved in the search.

$$\text{Precision ratio} = \frac{\text{Number of relevant document retrieved}}{\text{Total number of document retrieved in the search.}}$$

When multiplied by 100, we have the percentage

Example

You did a search; the database contains 50 numbers of relevant documents. The search produces for you 120 records. Out of 120 records, only 25 are relevant to your query. What is your recall and precision ratio?

$$\text{Recall ratio} = \frac{\text{number of relevant documents}}{\text{Total number of relevant document in the system}} = \frac{25}{50} = 0.5$$

Recall percentage= 0.5*100= 50%

$$\text{Precision ratio} = \frac{\text{number of relevant document retrieved}}{\text{Total number of document retrieved in the search.}} = \frac{25}{120} = 0.2$$

Precision percentage= 0.2*100= 20%

However, it must be noted that this measures must be related to requirement and tolerance of the user of the system in order to be meaningful. Other methods that the users employ in evaluating the output from a system are: Effort and Response time

Effort

Effort reflects the amount of input/effort expended by the user in examining the search products delivered to him. A user can expend effort in any of the following ways:

i. In specifying his information need

ii. In refining the search terms based on his initial output from the system

All these activities increase the user's interaction with the system, increase the amount of effort expended and ultimately yields better result from the system, the more likely that the user will rate the system performance as poor

Response Time

Response time measures the time from when the query is input to when the first set of items are retrieved for the user to use e.g goggle has a very fast response time that is why it has made it so popular

Point of view from the management: From the management point of view, **management evaluate** a system by the response time of the database and effort required to index an item on the database. The management are more concerned about the following:

1. The computer memory requirement for the system to perform at the expected level
2. Cost of hardware and software
3. Cost of training the personnel that will maintain the database and the hardware

From the management point of view, an efficient system is one that allows them to provide efficient services at the lowest cost and less time and less effort

In-text Question

Mention two main points of view in evaluating retrieval products

Answer

1. Point of view from the system user

2. Point of view from the management

2.2 Concept of Relevance

The most important evaluation metrics if an **information system** will always be biased by human subjectivity and this metric is the concept of **relevance**. **Relevance** is not always binary i.e relevant or irrelevant (yes/no).

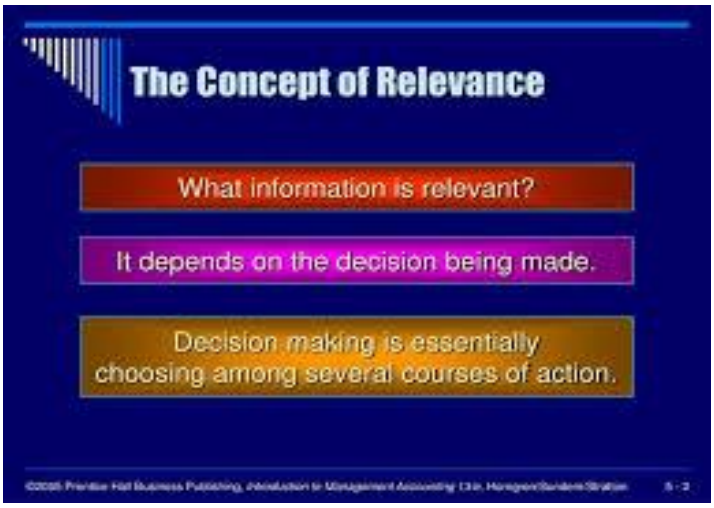

Fig 4.3.2: Concept of relevance

Relevance is a spectrum from an item being totally unrelated. **Relevance** may be:

1. Subjective depending on a user specific judgement
2. Situational related to user's requirement. Some users consider the information they already know to be irrelevant to their information need
3. Temporal- changing overtime because of more relevant information that is being find out
4. Measurable- observable at a point in time

The situation view of **relevance** pertains to the relationship between a user's information problem situation and the information needed to solve that problem. The situation view assumes that only user can make valid judgements regarding the suitability of information to solve their information need and this view is sometimes referred to as PERTINENCE where PERTINENCE is defined as those items that satisfy the information need at the time or point of retrieval.

The most common definition of **relevance** is known as Topical relevance i.e information items that meet the topic of the user's request or search

Situational relevance evaluates a document not only on the topic of the user's search but on its capability to provide information items that can be used by the user. Some researchers give the following criteria for regarding a document as

relevant

1. Topicality- the ability of the document to meet the topic of the search request
2. Perceived validity by the user- the extent to which the user perceives or judge the document to be valid
3. Novelty- the "newness" of the document to the user. In which case, the document already known by the user is considered irrelevant
4. Utility- the utility value of the document e.g. the information contains in the document open up a new research area for the user or was information enough for him to solve specific problem?
5. User's satisfaction- covers such factors as searchers confidence that all existing information on a topic has been retrieved

   Other measures of relevance

For a large database, it is not possible to know the total number of relevant document in the system, so researchers have suggested two approaches for knowing the total number:

1. Use sampling technique across the database and perform relevance judgement on the return items. The method gives an estimate of the total number of the relevant items in the database e.g. supposing a database contains 10,000 records. Taking a sample of every nth record and determine whether these records are relevant or not
2. Apply different search strategies to the same query. It is then assumed that all relevant documents in the database will be found from an aggregate of the searches. An extension of this is to combine results of several search engines to estimate the total number of relevant items

Aside recall and precision, there are other methods that users employ in evaluating the output

from a system. They are:

1. Unique Relevance Recall (URR): This is used to compare/measure two or more algorithms or systems. It measures the number of relevant items retrieved by one system or algorithm that were not retrieved by other

systems or algorithm.

So URR = $\frac{\text{number of unique relevant items}}{\text{Total number of relevant items}}$

Note: $\frac{\text{Total number of relevant items found by all algorithms}}{\text{Total number of unique items found by all algorithms}}$

2. Novelty ratio: This is a measure of newness. It is defined as the ratio of relevant items not previously known to the user to the total number of relevant items retrieved from the database.
3. Sought Recall: this is defined as the ratio of the total number of relevant items reviewed by the user after search to the total number of relevant items the user would have liked to examine

In-text Question

Itemize the criteria for regarding a document as relevant?

Answer

1. Topicality- the ability of the document to meet the topic of the search request

2. perceived validity by the user- the extent to which the user perceive or judge the document to be valid

3. Novelty- the "newness" of the document to the user. In which case, the document already known by the user is considered irrelevant

4. Utility- the utility value of the document e.g the information contain in the document open up a new research area for the user or was information enough for him to solve specific problem?

5. User's satisfaction- covers such factors as searchers confidence that all existing information on a topic has been retrieved

3.0 Study Session Summary and Conclusion

This study session explained how to evaluate retrieval products from two main points of views: system users and management. It also explains the concept of relevance by identifying criteria for regarding a document as relevant. Tutor marked assignment was given to test the students' ability to solve simple calculation in order to evaluate retrieval products. Self-Assessment questions were provided for students' practice.

4.0 Self-Assessment Questions

1. Enumerate and explainall the methods that users employ in evaluating the output from a system
2. Explain all the criteria for regarding a document as relevant

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2TxoAfk , https://bit.ly/2TvwqpO. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2HQ9Osj and critique it in the discussion forum

6.0 References/Further Readings

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Study Session 4

Standard Metadata and Their Description

Section and Subsection Headings:
Contents

1.0 Learning Outcomes
2.0 Main Contents
    2.1- What is Metadata?
    2.2- Functions of Metadata
    2.3- Creating Metadata
        2.3.1- Creation Tools
        2.3.2- Metadata Quality Control
    2.4- common Metadata Standards
3.0 Study Session Summary and Conclusion
4.0 Self-Assessment Questions and Answers
5.0 Additional Activities (Videos, Animations &Out of Class activities)
6.0 References/Further Readings

Introduction

You are welcome to study session 4, in this session, we shall consider what **metadata** is. You will also learn the functions of **metadata** as well as how to create **metadata.** Lastly, you will study common **metadata** standards.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Explain what metadata is and how to create it
2. Describe the functions of metadata
3. Discuss common metadata standards

2.0 Main Content

2.1 What is Metadata?

Metadata is structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource. It is the key to ensuring that resources will survive and continue to be accessible into the future. **Metadata** is often called data about data or information about information. A good example of **metadata** is the cataloguing system found in libraries, which records for example the author, title, subject, and location on the shelf of a resource.

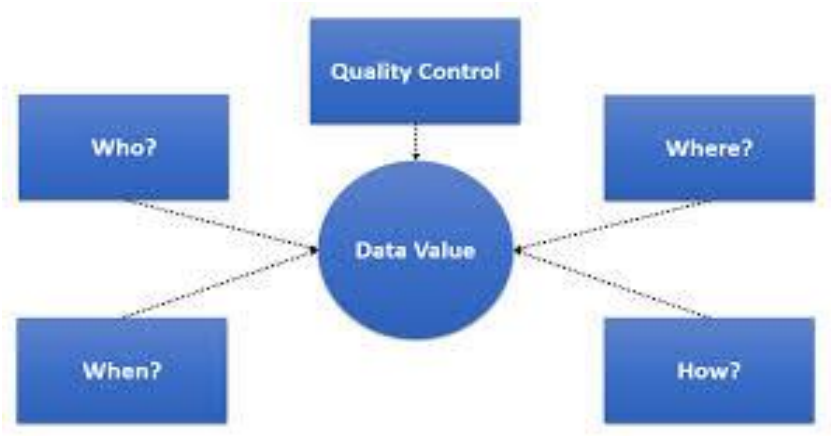

Fig 4.4.1: What is Metadata?

The term **metadata** is used differently in different communities. Some use it to refer to machine understandable information, while others use it only for records that describe electronic resources. In the library environment, metadata is commonly used for any formal scheme of resource description, applying to any type of object, digital or non- digital. Traditional library cataloguing is a form of **metadata**; MARC 21 and the rule sets used with it, such as AACR2, are metadata standards.

Other **metadata** schemes have been developed to describe various types of textual and non-textual objects including published books, electronic documents, archival finding aids, art objects, educational and training materials, and scientific **datasets**.

Let us look at the three main types of metadata:

1. *Descriptive metadata* - describes an information resource for purposes such as discovery, identification and retrieval. It can include elements such as title, abstract, author, and keywords.
2. *Structural metadata* - indicates how compound objects are put together or related to one another, for example, how pages are ordered to form chapters.

3. *Administrative metadata* - provides information to help manage an information resource, such as when and how it was created, file type and other technical information, and who can access it.

There are several subsets of administrative data. Two of them are sometimes listed as separate metadata types. These are:

1. *Rights management metadata*, which deals with intellectual property rights, and

2. *Preservation metadata*, which contains information needed to archive and preserve a resource.

2.2 Functions of Metadata

The main reasons for creating **metadata**, especially descriptive **metadata**, include: to facilitate discovery of relevant information, to aid resource discovery, help organise electronic resources, facilitate interoperability and legacy resource integration, provide digital identification and, support archiving and preservation. The functions are explained below

Resource Discovery

Metadata serves the same functions in resource discovery as good cataloguing does by:

1. Allowing resources to be found by relevant criteria
2. Identifying resources
3. Bringing similar resources together
4. Distinguishing dissimilar resources, and
5. Giving location information

Organising Electronic Resources

With the growth of web-based resources, aggregate websites become increasingly useful in organising links to resources based on audience or topic. Such lists can be built as static web pages, with the names and locations of the resources "**hardcoded**" in the HTML. The use of **metadata** stored in databases can allow these web pages to be built dynamically and more efficiently. Various software tools can be used to automatically extract and reformat the metadata information for web applications.

Interoperability

Describing a resource with **metadata** allows it to be understood by both humans and machines in ways that promote interoperability. Interoperability is the ability of multiple systems with different hardware and software platforms, data structures, and interfaces to exchange data with minimal loss of content and functionality. Using defined metadata schemes, shared transfer protocols, and crosswalks between schemes, resources across the network can be searched more seamlessly.

Two approaches to interoperability are cross-system search and metadata harvesting. The American National Standards Institute (ANSI) Z39.50 (a client-server protocol for searching and retrieving information from remote computer **databases**) protocol is commonly used for cross- system search. Z39.50 implementers do not share metadata but map their own search capabilities to a common set of search attributes. A contrasting approach taken by the Open Archives Initiative is for all data providers to translate their native metadata to a common core set of elements and expose this for harvesting.

A search service provider then gathers the **metadata** into a consistent central index to allow cross- repository searching regardless of the **metadata** formats used by participating repositories.

Digital Identification

Most **metadata** schemes include elements such as standard numbers to uniquely identify the work or object to which the **metadata** refers. The location of a digital object may also be given using a file name, URL (**Uniform Resource Locator**), or some more persistent identifier such as a PURL (Persistent URL) or DOI (**Digital Object Identifier**). Persistent identifiers are preferred because object locations often change, making the standard URL (**and therefore the metadata record**) invalid. In addition to the actual elements that point to the object, the metadata can be combined to act as a set of identifying data, differentiating one object from another for validation purposes.

Archiving and Preservation

Most current metadata efforts centre on the discovery of recently created resources. However, we observe that there is a growing concern that digital resources will not survive in usable form into the future. Digital information is fragile; it can be corrupted or altered, intentionally or unintentionally. It may become unusable as storage media and hardware and software technologies change. Format migration and perhaps emulation of current hardware and software behaviour in future hardware and software platforms are strategies for overcoming these challenges.

As we have mentioned above, ***metadata is the key to ensuring that resources will survive and continue to be accessible into the future*.** Archiving and preservation require special elements to track the lineage of a digital object (**where it came from and how it has changed over time**), to detail its physical characteristics, and to document its behaviour in order to emulate it on future technologies.

In-text Question

What is Metadata?

Answer

Metadata is structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource.

2.3 Creating Metadata

Who creates **metadata** depends on discipline, the information resource being described, the tools available, and the expected outcome. Though, it is almost always a cooperative effort. Much basic structural and administrative metadata is supplied by the technical staff, who initially digitises or otherwise create the digital object, or is generated through an automated process.

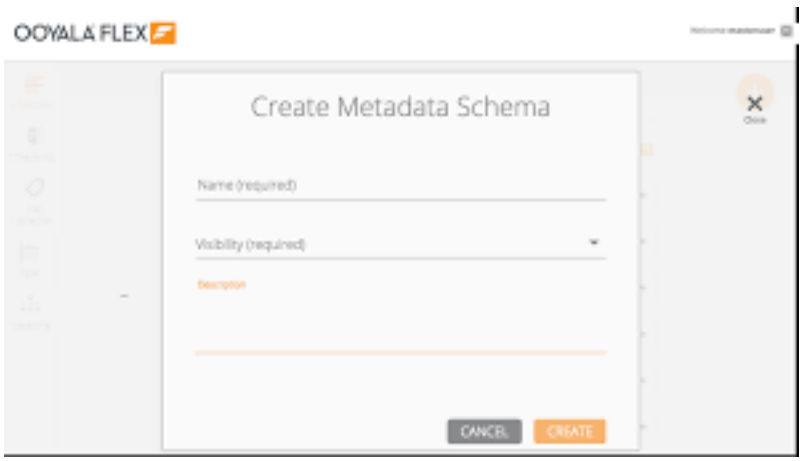

Fig 4.4.2: Creating Metadata

For descriptive **metadata**, it is best in some situations if the originator of the resource provides the information. This is particularly true in the documentation of scientific **datasets** where the originator has significant understanding of the rationale for the dataset and the uses to which it could be put, and for which there is little if any textual information from which an indexer could work.

However, we have found out that many projects have found that it is more efficient to have indexers or other information professionals create the descriptive **metadata**, because the authors or creators of the data do not have the time or the skills.

In other cases, a combination of researcher and information professional is used. The researcher may create a skeleton, completing the elements that can be supplied most readily. Then results may be supplemented or reviewed by the information specialist for consistency and compliance with the schema syntax and local guidelines

2.3.1 Creation Tools

A growing number of commercial software tools are becoming available. Again, many **metadata** project initiatives can develop tools and make them available to others, sometimes for free as is currently the situation. Creation tools fall into several categories:

1. *Templates:* allow a user to enter the metadata values into pre-set fields that match the element set being used. The template will then generate a formatted set of the element attributes and their corresponding values.
2. *Mark-up tools:* will structure the metadata attributes and values into the specified schema language. Most of these tools generate XML or SGML Document Type Definitions (DTD). Some templates include such a mark-up as part of their final translation of the metadata.
3. *Extraction tools:* will automatically createmetadata from an analysis of the digital information resource. These tools are generally limited to textual resources. The quality of the metadata extracted can vary significantly based on the tool's algorithms as well as the content and structure of the source text. These tools should be considered as an aid to creating metadata. The resulting metadata should always be manually reviewed and edited.
4. *Conversion tools:* will translate one metadata format to another. The similarity of elements in the source and target formats will affect how much additional editing and manual input of metadata may be required. Metadata tools are generally developed to support specific metadata schemas or element sets. The websites for the particular schema will frequently have links to relevant toolsets.

2.3.2 Metadata Quality Control

The creation of **metadata** automatically or by information originators who are not familiar with cataloguing, indexing, or vocabulary control can create quality problems. Mandatory elements may be missing or used incorrectly. Schema syntax may have errors that prevent the **metadata** from being processed correctly. Metadata content terminology may be inconsistent, making it difficult to locate relevant information.

The *Framework of Guidance for Building Good Digital Collections*, available on the National Information Standards Organisation (NISO) website, articulates six principles applying to good metadata:

1. Good metadata should be appropriate to the materials in the collection, users of the collection, and intended, current and likely use of the digital object.
2. Good metadata supports interoperability.
3. Good metadata uses standard controlled vocabularies to reflect the what, where, when and who of the content.
4. Good metadata includes a clear statement on the conditions and terms of use for the digital object.
5. Good metadata records are objects themselves and therefore should have the qualities of achievability, persistence, unique identification, etc. Good metadata should be authoritative and verifiable.
6. Good metadata supports the long-term management of objects in collections.

There are a number of ongoing efforts for dealing with the metadata quality challenge. These include:

1. Metadata creation tools are being improved with such features as templates, pick lists that limit the selection in a particular field, and improved validation rules.
2. Software interoperability programs that can automate the "crosswalk" between different schemas are continuously being developed and refined.
3. Content ~~orig~~inators are being formally trained in understanding metadata and controlled vocabulary concepts and in the use of metadata-related software tools.
4. Existing controlled vocabularies that may have initially been designed for a specific use or a narrow audience are getting broader use and awareness. For example, the *Content Types* and *Subtypes* originally defined for MIME email exchange are commonly used as the controlled list for the

Dublin Core *Format* element.

5. Communities of users are developing and refining audience- specific metadata schemas, application profiles, controlled vocabularies, and user guidelines. The *MODS User Guidelines* are a good example of the latter.

2.4 Common Metadata Standards

Metadata Standards are developed from schemes, which **metadata** elements grouped into sets designed for a specific purpose, example, for a specific domain or a particular type of information resource. For every element, the name and the semantics (**the meaning of the element**) are specified. Content rules (how content must be formulated), representation rules (e.g., **capitalisation rules**), and allowed element values (e.g., **from a controlled vocabulary**) can be specified optionally.

Some schemes also specify in which syntax the elements must be encoded, in contrast to syntax independent schemes. Many current schemes use Standards Generalised Mark-up Language (SGML) or XML to specify their syntax. Metadata schemes that are developed and maintained by standard organisations (**such as ISO**) or organisations that have taken on such responsibility (**such as the Dublin Core Metadata Initiative**) are called metadata standards.

Many different **metadata** schemes are being developed as standards across disciplines, such as library science, education, archiving, e- commerce, and arts. In the table below, an overview of some of the available metadata standards is given in the table below:

Table 4.4.1: Metadata Standard

Name	Focus	Description
EAD	Archiving	Encoded Archival Description - a standard for encoding archival finding aids using XML in archival and manuscript repositories.
CDWA	Arts	Categories for the Description of a Work of Art are a conceptual framework for describing and accessing information about works of art, architecture, and other material culture.
VRA Core	Arts	Visual Resource Association – the standard provides a categorical organization for the description of works of visual culture as well as the images that document them.
Darwin Core	Biology	The Darwin Core is a metadata specification for information about the geographic occurrence of species and the existence of specimens in collections.
ONIX	Book industry	Online information exchange- international standard for representing and communicating book industry product information in electronic form
CWM	Data warehousing	The main purpose of the common warehouse Metamodel is to enable easy interchange of warehouse and business intelligence metadata in distributed heterogeneous environment
IEEE LOM	Education	Learning objects Metadata – specifies the syntax and semantics of Learning Object Metadata
CSDGM	Geographic data	Content standard for Digital Geospatial Metadata maintained by the Federal Geographic Data Committee (FDGC)
e-GMS	Government	The e-Government Metadata standard defines the metadata elements for information resources to ensure maximum consistency of metadata across public sector organizations in UK.
GILS	Government/ Organizations	The Global Information Locator Service defines an open, lowcost, and scalable standard so that governments, companies and other organizations can help searchers find information
TEI	Humanities, social sciences and	Text Encoding Initiative- a standard foe the representation of texts in digital form chiefly in the humanities, social sciences and linguistics

	linguistics	
NISO MIX	Images	NISO Metadata for Images in XML is a XML schema for a set of technical data elements required to manage digital image collections
<indecs>	Intellectual properties	Interoperability of Data in E-Commerce Systems addresses the need to put different creation identifiers and metadata into a framework to support the management of intellectual property rights
MARC	Librarianship	Machine Readable Cataloguing - standards for the representation and communication of bibliographic and related information in machine-readable form.
METS	Librarianship	Metadata Encoding and Transmission Standard - standard for encoding descriptive, administrative, and structural metadata regarding objects within a digital library.
MODS	Librarianship	Metadata Object Description Standard - is a schema for a bibliographic element set that may be used for a variety of purposes, and particularly for library applications.
XOBIS	Librarianship	XML Organic Bibliographic Information Schema - a XML schema for modelling MARC data.
PBCore	Media	PBCore is a Metadata &Cataloguing Resource for Public Broadcasters & Associated Communities.
MPEG-7	Multimedia	The Multimedia Content Description Interface MPEG-7 is an ISO/IEC standard and specifies a set of descriptors to describe various types of multimedia information and is developed by the Moving Picture Experts Group.
DC	Networked resources	Dublin Core - interoperable online metadata standard focused on networked resources.
DOI	Networked resources	Digital Object Identifier - provides a system for the identification and hence management of information ("content") on digital networks, providing persistence and semantic interoperability.
ISO/IEC 11179[1]	Organizations	Standard that describes the metadata and activities needed to manage data elements in a registry to create a common understanding of data across organisational elements and between organisations.

ISO 15489[2]	Records management	Standard for records management policies and procedures.
moReq	Records management	A specification describing the ModelRequirements for the management of electronic records.
DIF	Scientific data sets	Directory Interchange Format - a descriptive and standardised format for exchanging information about scientific data sets.
SKOS	Web	Simple Knowledge Organisation System development by W3C of a simple yet powerful framework for expressing knowledge structures in a machine-understandable way, for use on the semantic web.

Source: Wikipedia, the free encyclopaedia, at
http://en.wikipedia.org/wiki/Metadata_standards

In-text Question
What are the six principles applying to good metadata?

Answer
1. Good metadata should be appropriate to the materials in the collection, users of the collection, and intended, current and likely use of the digital object.
2. Good metadata supports interoperability.
3. Good metadata uses standard controlled vocabularies to reflect the what, where, when and who of the content.
4. Good metadata includes a clear statement on the conditions and terms of use for the digital object.
5. Good metadata records are objects themselves and therefore should have the qualities of archivability, persistence, unique identification, etc. Good metadata should be authoritative and verifiable.
6. Good metadata supports the long-term management of objects in collections.

3.0 Study Session Summary and Conclusion

Metadata was defined as structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource. It is the key to ensuring that resources will survive and continue to be accessible into the future. You have studied standard **metadata** and their description and also learnt the functions of **metadata**, the tools for creating it and common **metadata**

standards.

4.0 Self-Assessment Questions

1. What is Metadata?
2. List and briefly discuss the functions of metadata.
3. Enumerate briefly on the metadata creation tools.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://bit.ly/2TxoAfk https://bit.ly/2TvwqpO. Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2HAn1Xy and critique it in the discussion forum

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Glossary

Boolean query
A query that is a Boolean combination of terms.

Classification
The process of deciding the appropriate category for a given document. Examples are deciding what newsgroup an article belongs in, what folder an email message should be directed to, or what is the general topic of an essay.

Cluster
A grouping of representations of similar documents. In a vector space model, one can perform retrieval by comparing a query vector with the centroids of clusters. One can continue search in those clusters that are in this way most promising.

Collaborative Filtering
The process of filtering documents by determining what documents other users with similar interests and/or needs found relevant. Also called "social filtering".

Collection
A group of documents that a user wishes to get information from. See also test collection.

Collection Fusion
The problem of combining the search results from multiple collections. This could be tricky since some measures such as IDF will differ across collections, and, if one retrieves a fixed number of documents, it is unclear how many to take from each collection.

Content-Based Filtering
The process of filtering by extracting features from the text of documents to determine the documents' relevance. Also called "cognitive filtering".

Document
A piece of information the user may want to retrieve. This could be a text file, a WWW page, a newsgroup posting, a picture, or a sentence from a book.

Indexing
The process of converting a collection into a form suitable for easy search and retrieval.

Information Extraction
A related area that attempts to identify semantic structure and other specific types of information from unrestricted text.

Information Filtering
Given a large amount of data, return the data that the user wants to see. This is the standard problem in IR.

Information Need
What the user really wants to know. A query is an approximation to the information need.

Information Retrieval
The study of systems for indexing, searching, and recalling data, particularly text or other unstructured forms.

Inverse Document Frequency
Abbreviated as IDF, this is a measure of how often a particular term appears across all of the documents in a collection. It is usually defined as log(collection size/number of documents containing the term). So common words will have low

IDF and words unique to a document will have high IDF. This is typically used for weighting the parameters of a model.

Inverted File

A representation for a collection that is essentially an index. For each word or term that appears in the collection, an inverted file lists each document where it appears. This representation is especially useful for performing Boolean queries.

Precision

A standard measure of IR performance, precision is defined as the number of relevant documents retrieved divided by the total number of documents retrieved.

Precoordination of terms

The process of using compound terms to describe a document. For example, this page may be indexed under the term "information retrieval glossary".

Postcoordination of terms

The process of using single terms to describe a document which are then combined (or coordinated) based on a given query. For example, this page may be indexed under the words INFORMATION, RETRIEVAL, and GLOSSARY. We'd then have to combine these terms based on a query like "INFORMATION and RETRIEVAL".

Probabilistic Model

Any model that considers the probability that a term or concept appears in a document, or that a document satisfies the information need. A Bayesian inference net is a good framework for this style of model. The INQUERY system is the most successful example.

Query

A string of words that characterizes the information that the user seeks. Note that this does not have to be an English language question.

Query Expansion

Any process which builds a new query from an old one. It could be created by adding terms from other documents, as in relevance feedback, or by adding synonyms of terms in the query (as found in a thesaurus).

Question Answering

The problem of finding the exact answer to a user's natural language question in a large collection.

Recall

A standard measure of IR performance, recall is defined as the number of relevant documents retrieved divided by the total number of relevant documents in the collection. For example, suppose there are 80 documents relevant to widgets in the collection. System X returns 60 documents, 40 of which are about widgets. Then X's recall is 40/80 = 50%. In an ideal world, recall is 100%. However, since this is trivial to achieve (by retrieving *all* of the documents), a system attempts to maximize both recall and precision simultaneously.

Relevance

An abstract measure of how well a document satisfies the user's information need. Ideally, your system should retrieve all of the relevant documents for you. Unfortunately, this is a subjective notion and difficult to quantify.

Relevance Feedback

A process of refining the results of a retrieval using a given query. The user indicates which documents from those returned are most relevant to his query. The system typically tries to find terms common to that subset, and adds them to

the old query. It then returns more documents using the revised query. This can be repeated as often as desired. Also called "find similar documents" or "query by example".

Robot
See spider.

Routing
Similar to information filtering, the problem of retrieving wanted data from a continuous stream of incoming information (i.e. long-term filtering).

SIGIR
The ACM's special interest group on Information Retrieval. They publish SIGIR Forum and have an annual conference. For more information, check their home page.

Signature File
A representation of a collection where documents are hashed to a bit string. This is essentially a compression technique to permit faster searching.

Similarity
The measure of how alike two documents are, or how alike a document and a query are. In a vector space model, this is usually interpreted as how close their corresponding vector representations are to each other. A popular method is to compute the cosine of the angle between the vectors.

Spider
Also called a robot, a program that scans the web looking for URLs. It is started at a particular web page, and then access all the links from it. In this manner, it traverses the graph formed by the WWW. It can record information about those servers for the creation of an index or search facility. Most search engines are created using spiders. The problem with them is, if not written properly, they can

make a large number of hits on a server in a short space of time, causing the system's performance to decay. For more information, look here.

Stemming

The process of removing prefixes and suffixes from words in a document or query in the formation of terms in the system's internal model. This is done to group words that have the same conceptual meaning, such as WALK, WALKED, WALKER, and WALKING. Hence the user doesn't have to be so specific in a query.

Stopword

A word such as a preposition or article that has little semantic content. It also refers to words that have a high frequency across a collection. Since stopwords appear in many documents, and are thus not helpful for retrieval, these terms are usually removed from the internal model of a document or query.

Term

A single word or concept that occurs in a model for a document or query. It can also refer to words in the original text.

Term Frequency

Abbreviated as TF, the number of times a particular term occurs in a given document or query. This count is used in weighting the parameters of a model.

Test collection

A collection specifically created for evaluating experimental IR systems. It usually comes with a set of queries, and a labelling (decided by human experts) that decides which documents are relevant to each query. TIPSTER is one of the most prevalent test collections currently. Another useful collection for classification is the Reuters text categorization test collection. Here there are

no queries, but the documents are news articles labelled with a variety of topic designations.

TIPSTER

An ongoing project where various groups and institutions have pooled their resources to solve problems in routing and information extraction. The framework is such that each team can work on a different piece and simply "plug" their application into the general architecture. The project also has a large test collection available.

TREC

Text REtrieval Conference. This group gives IR researchers a common test collection and a common evaluation system. Hence, systems can be compared and contrasted on the same data. You can visit the conference's home page for information about the conference and on-line versions of the proceedings.

Vector Space Model

A representation of documents and queries where they are converted into vectors. The features of these vectors are usually words in the document or query, after stemming and removing stopwords. The vectors are weighted to give emphasis to terms that exemplify meaning, and are useful in retrieval. In a retrieval, the query vector is compared to each document vector. Those that are closest to the query are considered to be similar, and are returned. SMART is the most famous example of a system that uses a vector space model.

Weighting

Usually referring to terms, the process of giving emphasis to the parameters for more important terms. In a vector space model, this is applied to the features of each vector. A popular weighting scheme is TF*IDF. Other possible schemes are Boolean (1 if the term appears, 0 if not), or by term frequency alone. In a vector model, the weights are sometimes normalized to sum to 1, or by dividing by the square root of the sum of their squares.